

中国科学院大学

University of Chinese Academy of Sciences

Generative Models & AI4S Applications

Lecture 6 Variational Autoencoder II

Fall 2026

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Lecture Outline

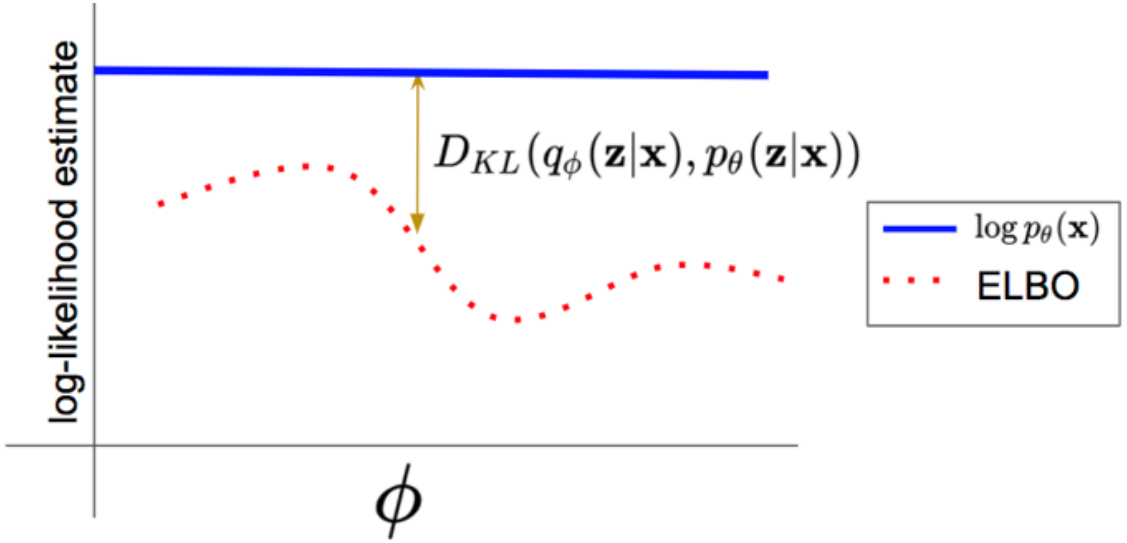
- Variational Learning
- Variational Autoencoder (VAE)
- Re-parameterization Trick
- The Generative Latent Space of VAE
- Advanced Variants of VAE

Lecture Outline

- **Variational Learning**
- **Variational Autoencoder (VAE)**
- **Re-parameterization Trick**
- **The Generative Latent Space of VAE**
- **Advanced Variants of VAE**

Variational Learning

- **ELBO and Variational Inference**
  - **Goal:** Maximize the ELBO over dataset $\mathcal{D}$ by **jointly optimizing θ and ϕ**



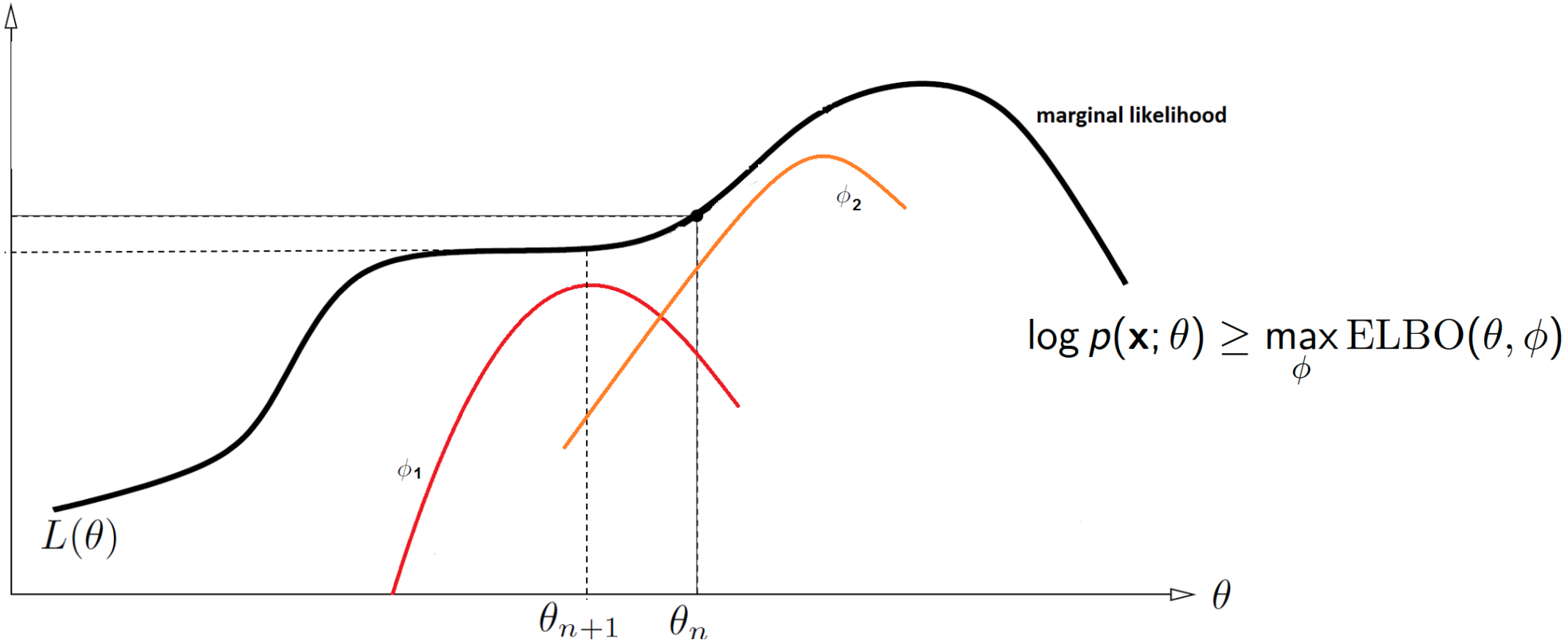
  - **Optimize ϕ:** Tighten the bound by making $q_\phi(z \mid x)$ closer to $p_\theta(z \mid x)$

$$\log p(\mathbf{x}; \theta) \geq \max_{\phi} \mathrm{ELBO}(\theta, \phi)$$

  - **Optimize θ:** Train with tight ELBO as a **tractable surrogate** for $\log p_\theta(x)$

Variational Learning

- **ELBO and Variational Inference**
  - **Goal:** Maximize the ELBO over dataset $\mathcal{D}$ by **jointly optimizing θ and ϕ**



$\mathcal{L}(x;\theta,\phi_1)$ and $\mathcal{L}(x;\theta,\phi_2)$ are both lower bounds (ELBOs)

Variational Learning

- **Variational Learning on the Training Set $\mathcal{D}$**
  - The ELBO holds for **any sample x** and **any $q(z;\phi)$**

$$\log p(\mathbf{x};\theta) \geq \underbrace{\mathcal{L}(\mathbf{x};\theta,\phi)}_{\text{ELBO}}$$

  - The log-likelihood $\log p(x;\theta)$ over the entire dataset $\mathcal{D} = \{x^1, \ldots, x^M\}$ is

$$\ell(\theta;\mathcal{D}) = \sum_{\mathbf{x}^i \in \mathcal{D}} \log p(\mathbf{x}^i;\theta) \geq \sum_{\mathbf{x}^i \in \mathcal{D}} \mathcal{L}(\mathbf{x}^i;\theta,\phi^i)$$

  - **Challenge:** True posterior $p(z \mid x^i;\theta)$ is different across data points x^i

    ➔ Different ϕ^i needs to be optimized for every x^i

$$\ell(\theta;\mathcal{D}) \geq \max_{\phi^1,\cdots,\phi^M} \sum_{\mathbf{x}^i \in \mathcal{D}} \mathcal{L}(\mathbf{x}^i;\theta,\phi^i)$$

Hard to scale to large dataset !

Variational Learning

- **Amortized Inference**
  - Optimizing ϕ^i for each $\boldsymbol{x}^i$ in $\mathcal{D}$ is expensive

$$\ell(\theta; \mathcal{D}) \geq \max_{\phi^1, \cdots, \phi^M} \sum_{\mathbf{x}^i \in \mathcal{D}} \mathcal{L}(\mathbf{x}^i; \theta, \phi^i)$$

  - **Amortization (摊销):** "Reducing cost by repeated re-use"

    ➔ Replace per-sample optimization with a **shared** inference network

    ✓ Instead of optimizing one $\phi^{i,*}$ for each x^i

$$x^i \mapsto \phi^{i,*}$$

    ✓ Learn one shared encoder $q_\phi(z \mid x)$ that outputs the approximate posterior for any x^i (i.e., **Amortized Inference**)

Variational Learning

- **Variational Learning with Amortized Inference**
  - ➢ **Training Procedure for Latent Variable Models**

$$\max_{\theta} \ell(\theta; \mathcal{D}) \geq \max_{\theta,\phi} \sum_{\mathbf{x}^i \in \mathcal{D}} \mathcal{L}(\mathbf{x}^i; \theta, \phi)$$

Shared parameters for ϕ

where $\mathcal{L}(\mathbf{x}; \theta, \phi) = \underbrace{E_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p(\mathbf{z}, \mathbf{x}; \theta) - \log q_\phi(\mathbf{z}|\mathbf{x})]}_{\text{ELBO}(\boldsymbol{\theta}, \boldsymbol{\phi})}$

Variational Learning

- **Variational Learning with Amortized Inference**
  - **Training Procedure for Latent Variable Models**

$$\max_{\theta} \ell(\theta; \mathcal{D}) \geq \max_{\theta,\phi} \sum_{\mathbf{x}^i \in \mathcal{D}} \mathcal{L}(\mathbf{x}^i; \theta, \phi)$$

where $\mathcal{L}(\mathbf{x}; \theta, \phi) = E_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p(\mathbf{z}, \mathbf{x}; \theta) - \log q_\phi(\mathbf{z}|\mathbf{x})]$

① Initialize $\theta^{(0)}$, $\phi^{(0)}$;

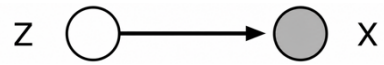
② Randomly sample a data point x^i from $\mathcal{D}$

③ Compute $\nabla_\theta \mathcal{L}(x^i; \theta, \phi)$ and $\nabla_\phi \mathcal{L}(x^i; \theta, \phi)$

④ Update θ and ϕ in the gradient direction

Summary: Latent Variable Model

- **Modeling:** $p(x, z) = p(x \mid z)p(z)$, z is the latent variable



- **Training: Maximum likelihood**
  - **Importance Sampling**

Monte Carlo Estimation

$$p_\theta(\mathbf{x}) = \sum_{\text{All values of } \mathbf{z}} p_\theta(\mathbf{x}, \mathbf{z}) = \sum_{\mathbf{z} \in \mathcal{Z}} \frac{q(\mathbf{z})}{q(\mathbf{z})} p_\theta(\mathbf{x}, \mathbf{z}) = \mathbb{E}_{\mathbf{z} \sim q(\mathbf{z})} \left[\frac{p_\theta(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} \right] \Rightarrow p_\theta(\mathbf{x}) \approx \frac{1}{k} \sum_{j=1}^{k} \frac{p_\theta(\mathbf{x}, \mathbf{z}^{(j)})}{q(\mathbf{z}^{(j)})}$$

  - **Jensen's Inequality**

$$\log p(\mathbf{x}; \theta) = \log \left(\sum_{\mathbf{z} \in \mathcal{Z}} p_\theta(\mathbf{x}, \mathbf{z}) \right) = \log \left(\mathbb{E}_{\mathbf{z} \sim q(\mathbf{z})} \left[\frac{p_\theta(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} \right] \right) \geq \mathbb{E}_{\mathbf{z} \sim q(\mathbf{z})} \left[\log \left(\frac{p_\theta(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} \right) \right]$$

Evidence Lower Bound (ELBO)

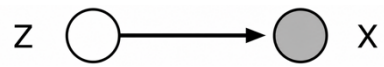
  - **Equality Condition:** $q(z) = p(z \mid x; \theta)$

$$\log p(\mathbf{x}; \theta) = \text{ELBO} + D_{KL}(q(\mathbf{z}) \| p(\mathbf{z} | \mathbf{x}; \theta))$$

Gap determined by $q(z)$

Summary: Latent Variable Model

- **Modeling:** $p(x, z) = p(x \mid z)p(z)$, z is the latent variable

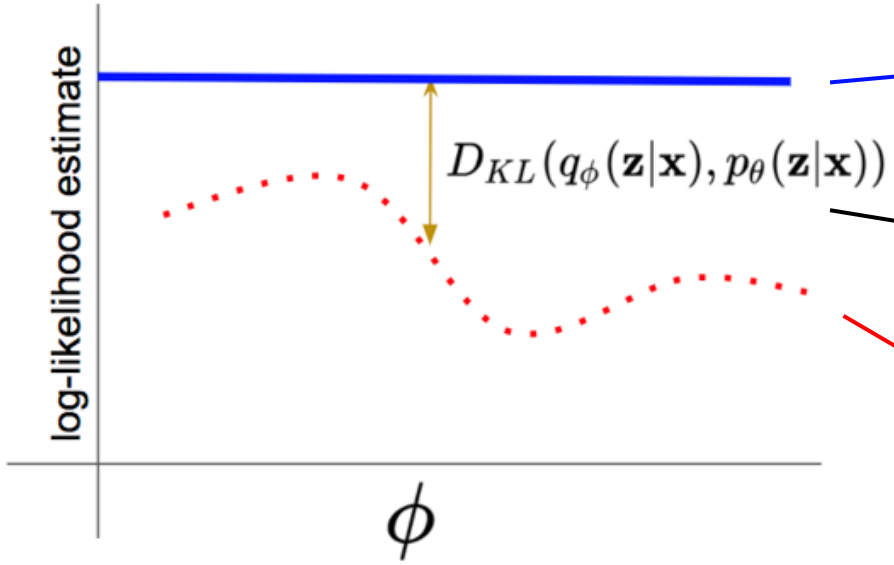


- **Training: Maximum likelihood**

$$\log p(\mathbf{x}; \theta) = \mathrm{ELBO} + D_{KL}(q(\mathbf{z}) \| p(\mathbf{z}|\mathbf{x}; \theta))$$

  - **Variational Inference (变分推断)**

$$q(z) \coloneqq q(z; \phi) \rightarrow p_\theta(z \mid x; \theta)$$



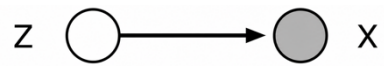
$\log p(\mathbf{x}; \theta)$: Target, but **intractable**

$D_{KL}(q(\mathbf{z}) \| p(\mathbf{z}|\mathbf{x}; \theta))$: Error if approximating $p(z \mid x; \theta)$ with $q(z)$

ELBO : Lower bound, and **tractable**

Summary: Latent Variable Model

- **Modeling:** $p(x, z) = p(x \mid z)p(z)$, z is the latent variable



- **Training: Maximum likelihood**

$$\log p(\mathbf{x}; \theta) = \mathrm{ELBO} + D_{KL}(q(\mathbf{z}) \| p(\mathbf{z}|\mathbf{x}; \theta))$$

  - **Variational Inference (变分推断)**

$$q(z) \coloneqq q(z; \phi) \rightarrow p_\theta(z \mid x; \theta)$$

⇒ True posterior $p(z \mid x^i; \theta)$ is different across x^i

⇓

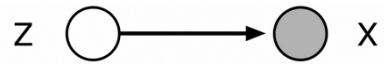
  - **Amortization (摊销):**

$$q(z; \phi^i) \rightarrow q_\phi(z \mid x)$$

⇐ Different ϕ^i needs to be optimized for every x^i

Summary: Latent Variable Model

- **Modeling:** $p(x, z) = p(x \mid z)p(z)$, z is the latent variable



- **Training: Maximum likelihood**

$$\log p(\mathbf{x}; \theta) = \mathrm{ELBO} + D_{KL}(q(\mathbf{z}) \| p(\mathbf{z}|\mathbf{x}; \theta))$$

  - **Variational Inference (变分推断)**

$$q(z) \coloneqq q(z; \phi) \rightarrow p_\theta(z \mid x; \theta)$$

  - **Amortization (摊销):**

$$q(z; \phi^i) \rightarrow q_\phi(z \mid x)$$

  - **Variational Learning on Dataset $\mathcal{D}$:**

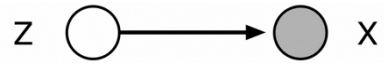
Log-likelihood (θ) ⇒ $\max_\theta \ell(\theta; \mathcal{D}) \geq \max_{\theta,\phi} \sum_{\mathbf{x}^i \in \mathcal{D}} \mathcal{L}(\mathbf{x}^i; \theta, \phi)$ ⇐ **ELBO** (θ, ϕ)

$$\mathcal{L}(\mathbf{x}; \theta, \phi) = E_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p(\mathbf{z}, \mathbf{x}; \theta) - \log q_\phi(\mathbf{z}|\mathbf{x}))]$$

$$\mathbb{E}_{\mathbf{z} \sim q(\mathbf{z})}\left[\log\left(\frac{p_\theta(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})}\right)\right]$$

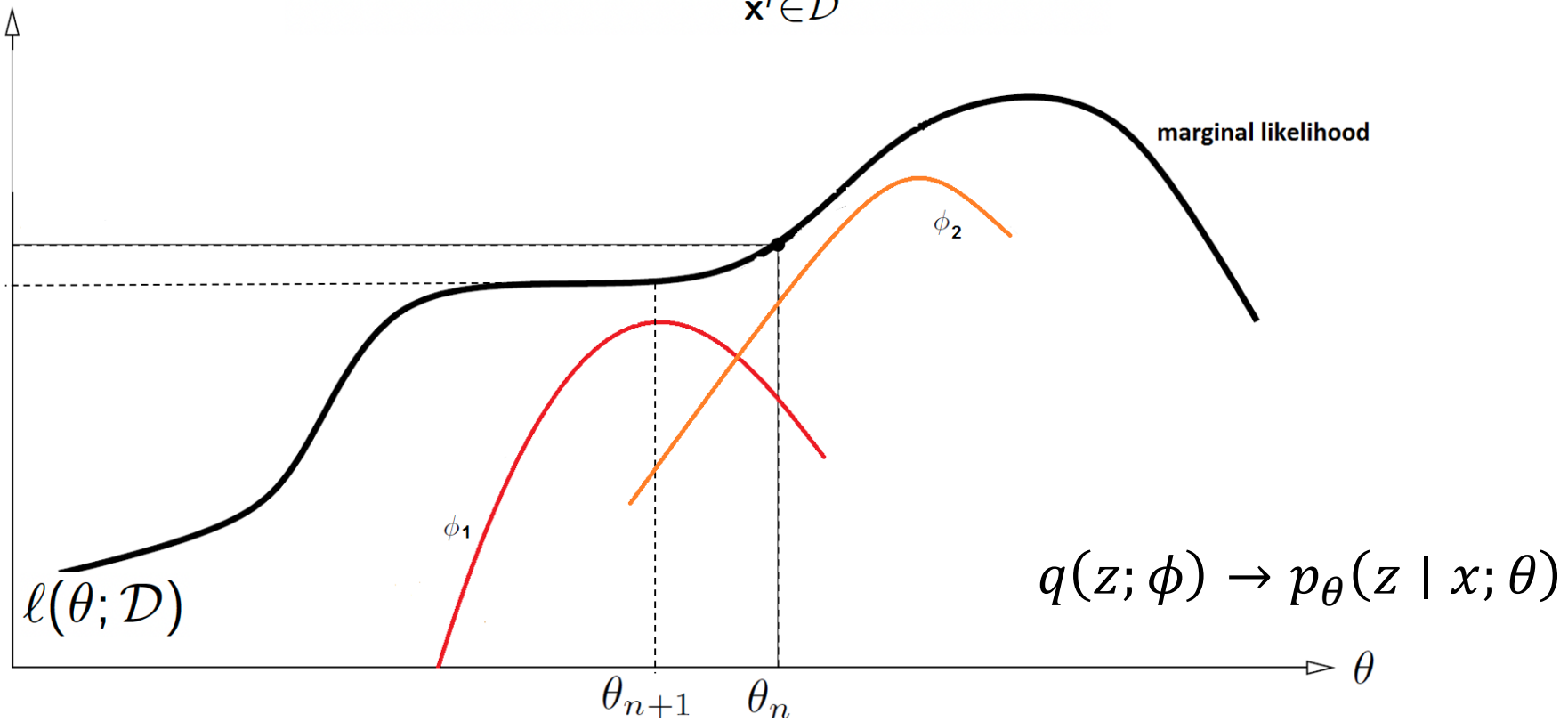
Summary: Latent Variable Model

- **Modeling:** $p(x, z) = p(x \mid z)p(z)$, z is the latent variable



  - ➢ **Variational Learning on Dataset $\mathcal{D}$:**

Log-likelihood (θ) ⇒ $\max_{\theta} \ell(\theta; \mathcal{D}) \geq \max_{\theta,\phi} \sum_{\mathbf{x}^i \in \mathcal{D}} \mathcal{L}(\mathbf{x}^i; \theta, \phi)$ ⇐ **ELBO** (θ, ϕ)

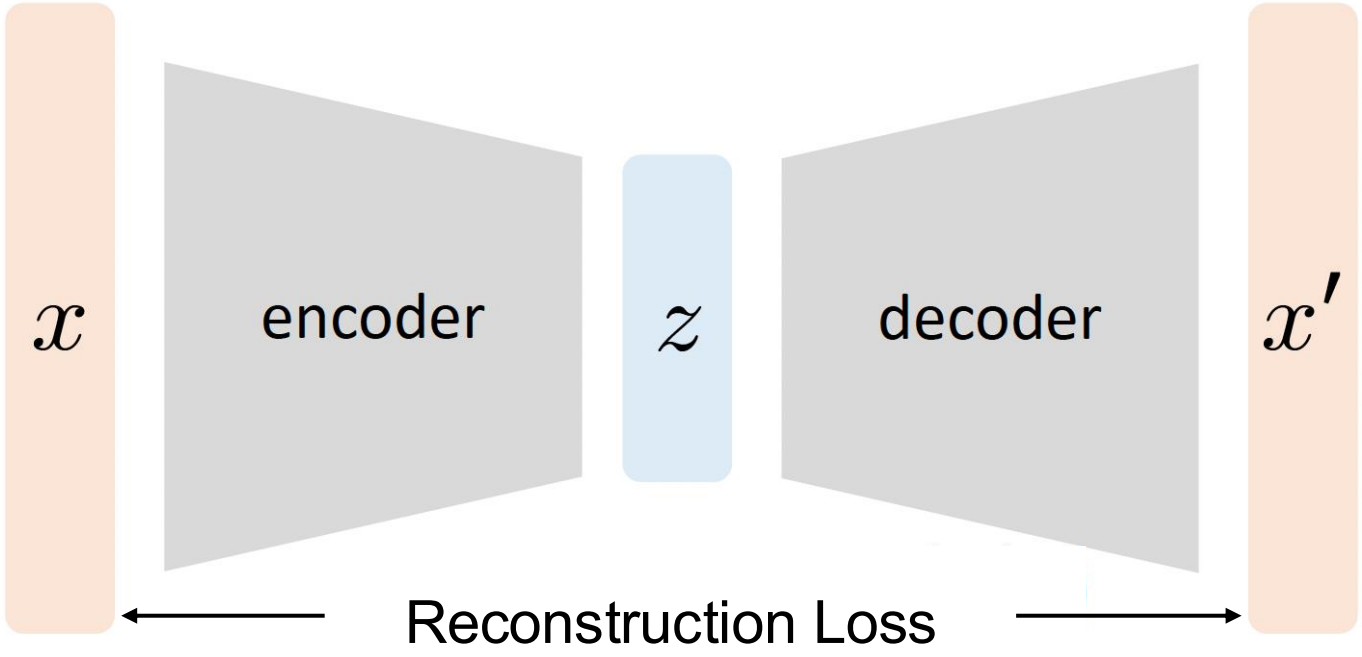


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Variational Autoencoder

- **Latent Variable Model: An Autoencoder Perspective**

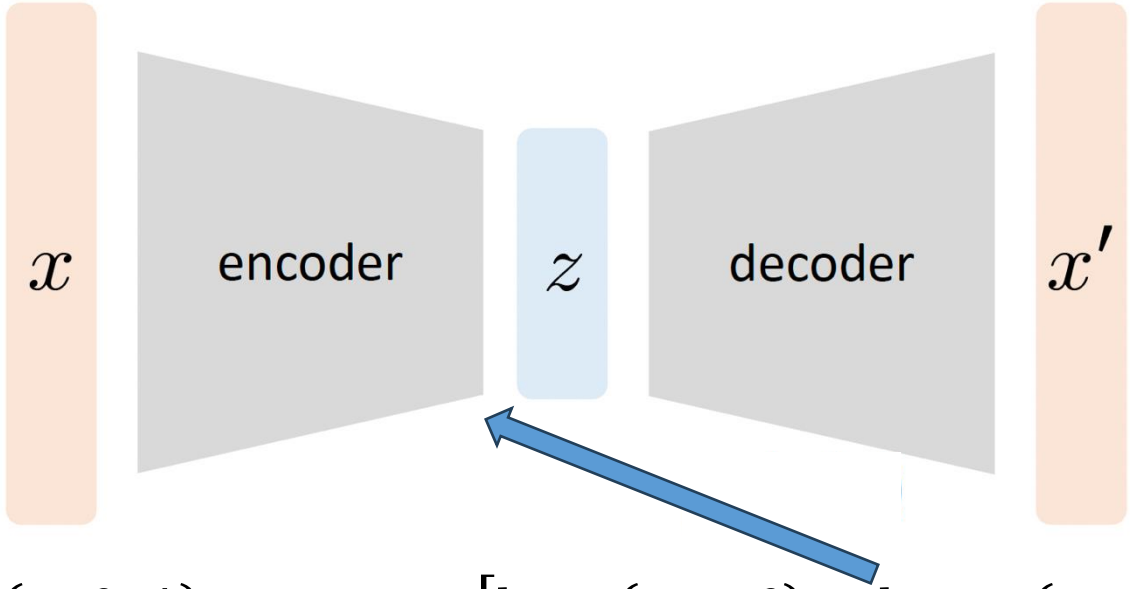


$$\max_{\theta} \ell(\theta; \mathcal{D}) \geq \max_{\theta,\phi} \sum_{\mathbf{x}^i \in \mathcal{D}} \mathcal{L}(\mathbf{x}^i; \theta, \phi)$$

ELBO (θ, ϕ) $\mathcal{L}(\mathbf{x}; \theta, \phi) = E_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p(\mathbf{z}, \mathbf{x}; \theta) - \log q_\phi(\mathbf{z}|\mathbf{x}))]$

Variational Autoencoder

- **Latent Variable Model: An Autoencoder Perspective**

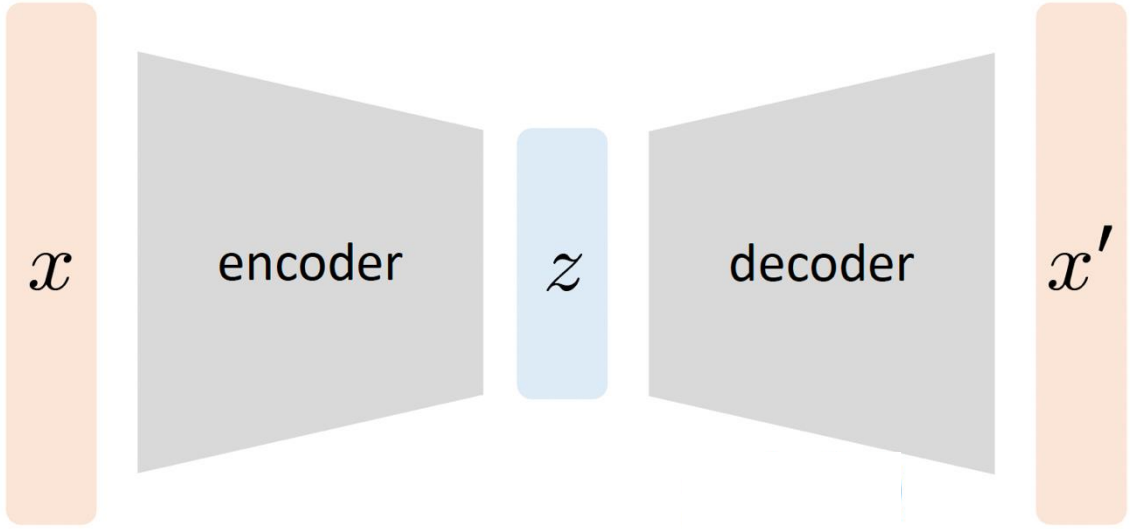


ELBO (θ, ϕ) $\quad \mathcal{L}(x; \theta, \phi) = \mathbb{E}_{q_\phi(z|x)}\left[\log p(z, x; \theta) - \log q_\phi(z \mid x)\right]$

$p(x \mid z; \theta)$**?**

Variational Autoencoder

- **Latent Variable Model: An Autoencoder Perspective**



ELBO (θ, ϕ)

$$\mathcal{L}(x;\theta,\phi) = \mathbb{E}_{q_\phi(z|x)}\left[\log p(z,x;\theta) - \log q_\phi(z \mid x)\right]$$

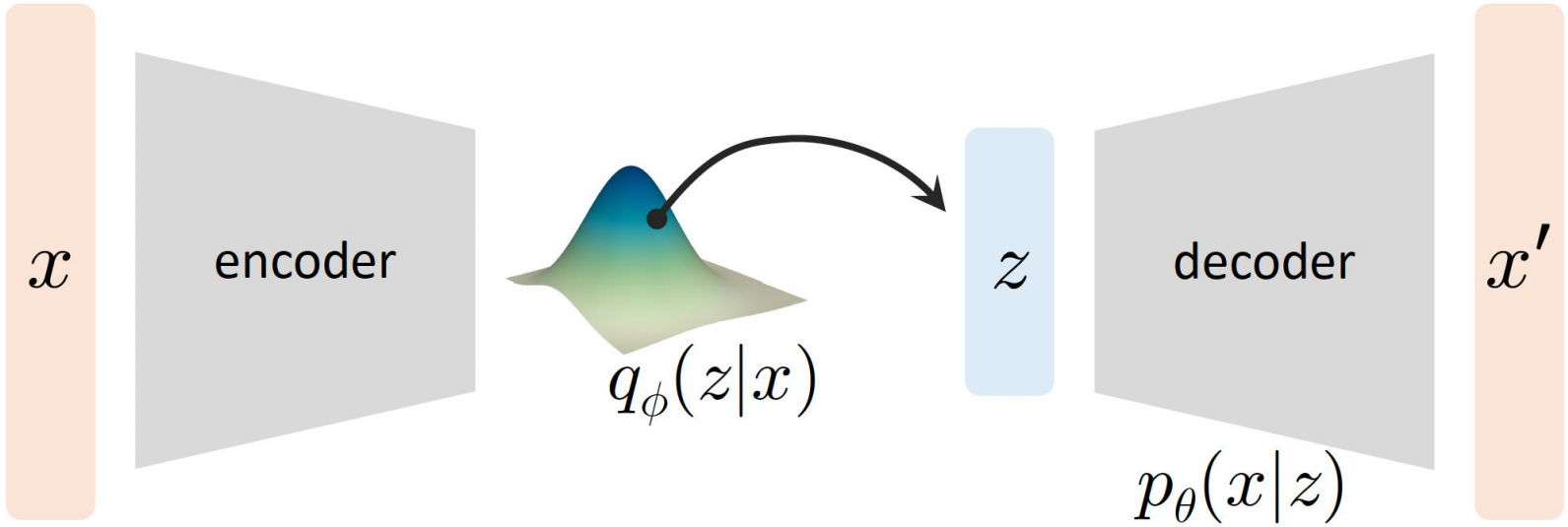
$$= \mathbb{E}_{q_\phi(z|x)}\left[\log\left(p(x \mid z;\theta)\cdot p(z)\right) - \log q_\phi(z \mid x)\right]$$

$$= \mathbb{E}_{q_\phi(z|x)}\left[\log p(x \mid z;\theta) + \log p(z) - \log q_\phi(z \mid x)\right]$$

$$= \mathbb{E}_{q_\phi(z|x)}\left[\log p(x \mid z;\theta)\right] - D_{KL}\left(q_\phi(z \mid x) \parallel p(z)\right)$$

Variational Autoencoder

- **Implementation: Neural Network**

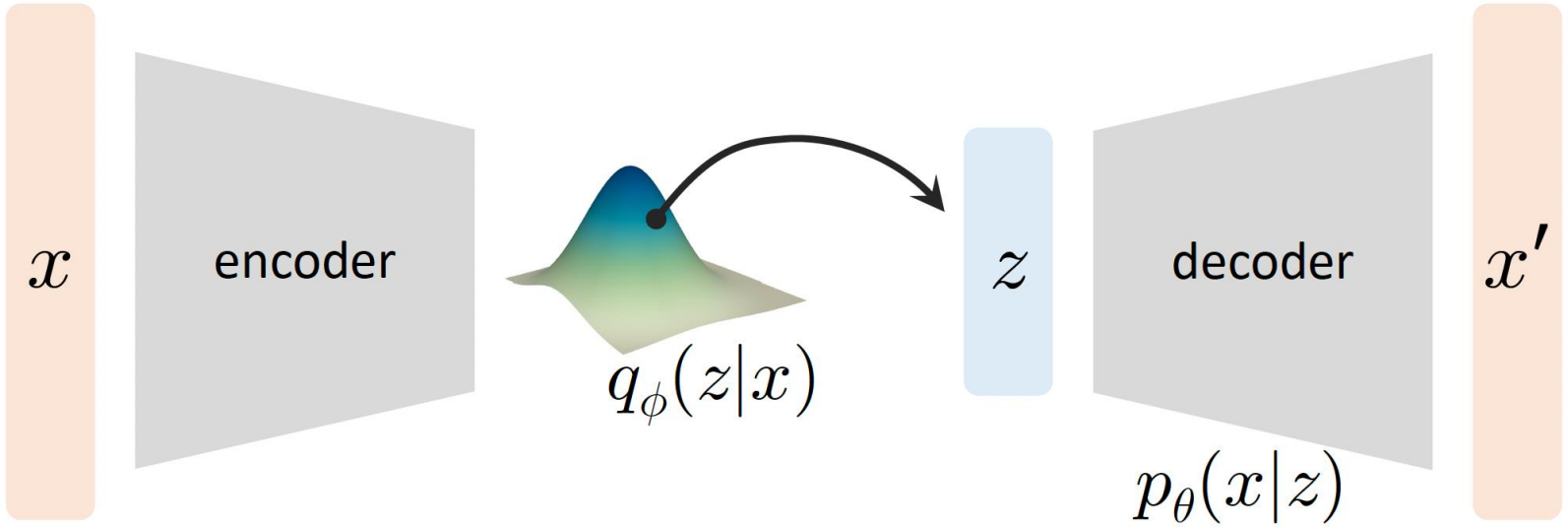


ELBO (θ, ϕ) $\mathcal{L}(x;\theta,\phi) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - D_{KL}\left(q_\phi(z \mid x) \parallel p(z)\right)$

- **Question:** How to connect networks with probabilistic distributions?
  - ✓ Encoder $f_\phi(x)$ ➔ $q_\phi(z \mid x)$
  - ✓ Decoder $g_\theta(z)$ ➔ $p_\theta(x \mid z)$

Variational Autoencoder

- **Implementation: Neural Network**



ELBO (θ, ϕ) $\mathcal{L}(x;\theta,\phi) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - D_{KL}\left(q_\phi(z \mid x) \parallel p(z)\right)$

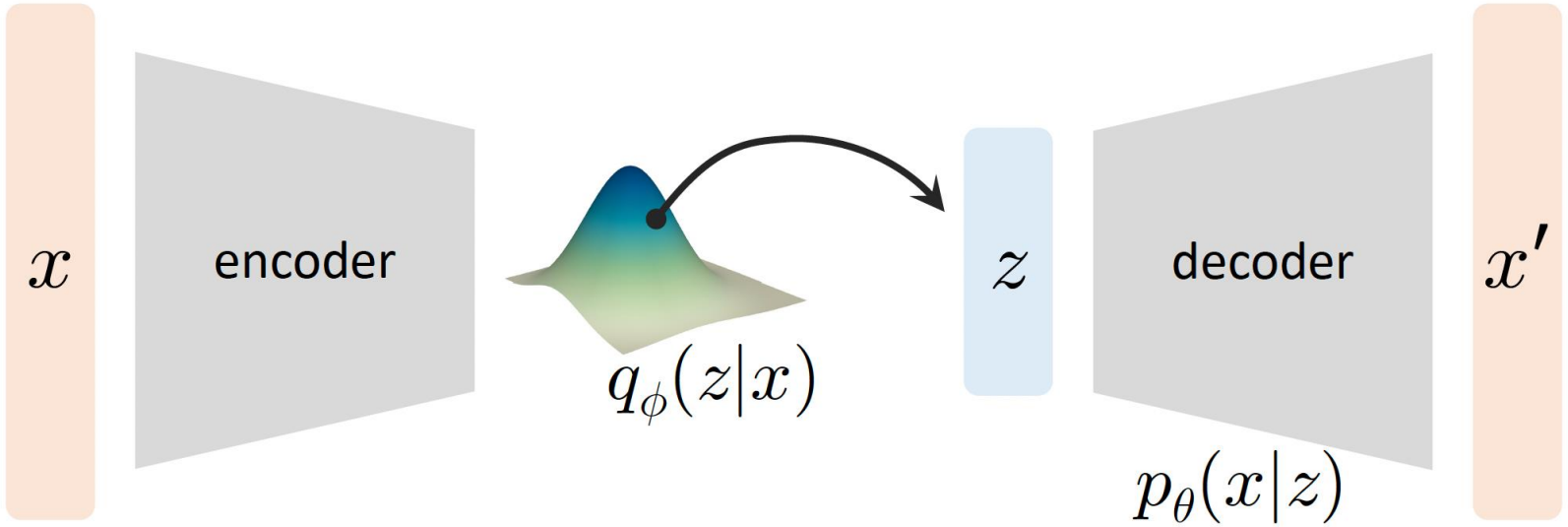
➢ **Question:** How to connect networks with probabilistic distributions?

✓ Distribution: $z \sim q_\phi(z \mid x) = \mathcal{N}\left(z; \mu(x), \sigma(x)\right)$

✓ Encoder Network: $f_\phi(x) \rightarrow \left(\mu(x), \sigma(x)\right)$

Variational Autoencoder

- **Implementation: Neural Network**

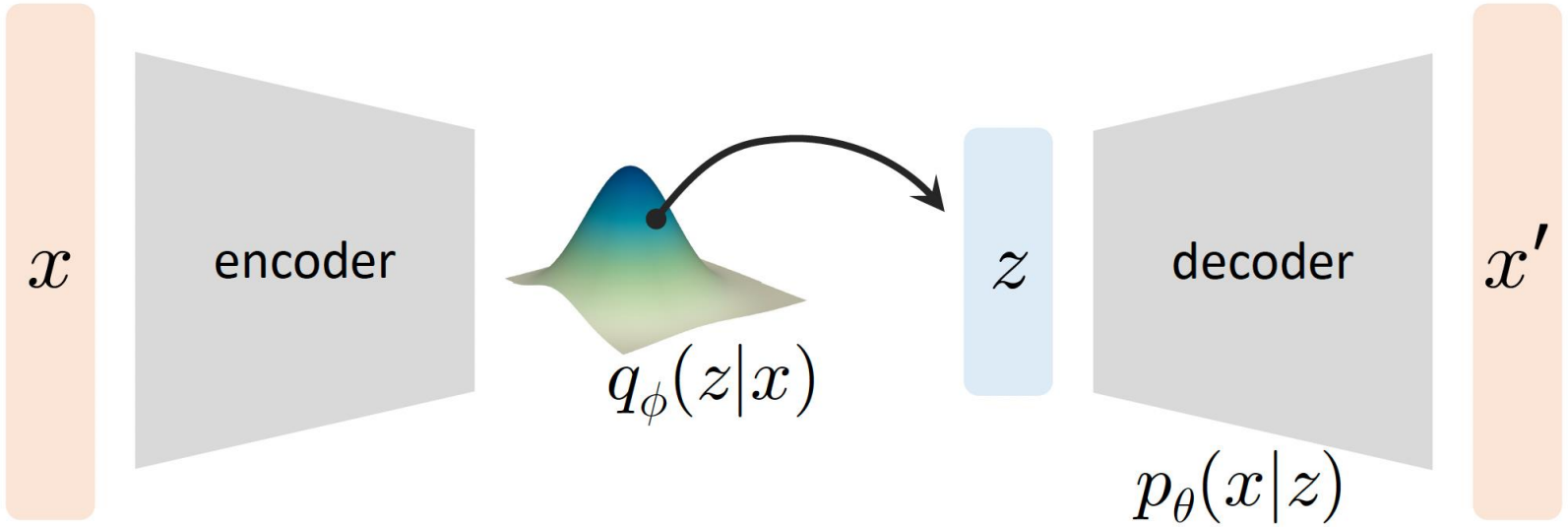


ELBO (θ, ϕ) $\mathcal{L}(x;\theta,\phi) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - D_{KL}\left(q_\phi(z \mid x) \parallel p(z)\right)$

- **Question:** How to connect networks with probabilistic distributions?
  - ✓ Distribution: $p_\theta(x \mid z) \sim \mathcal{N}(x; x'(z), \sigma_0^2 I)$, assuming σ_0 is known
  - ✓ Decoder Network: $g_\theta(z) \to x'(z)$

Variational Autoencoder

- **Implementation: Objective Functions**



ELBO (θ, ϕ) $\mathcal{L}(x;\theta,\phi) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - D_{KL}\left(q_\phi(z \mid x) \parallel p(z)\right)$

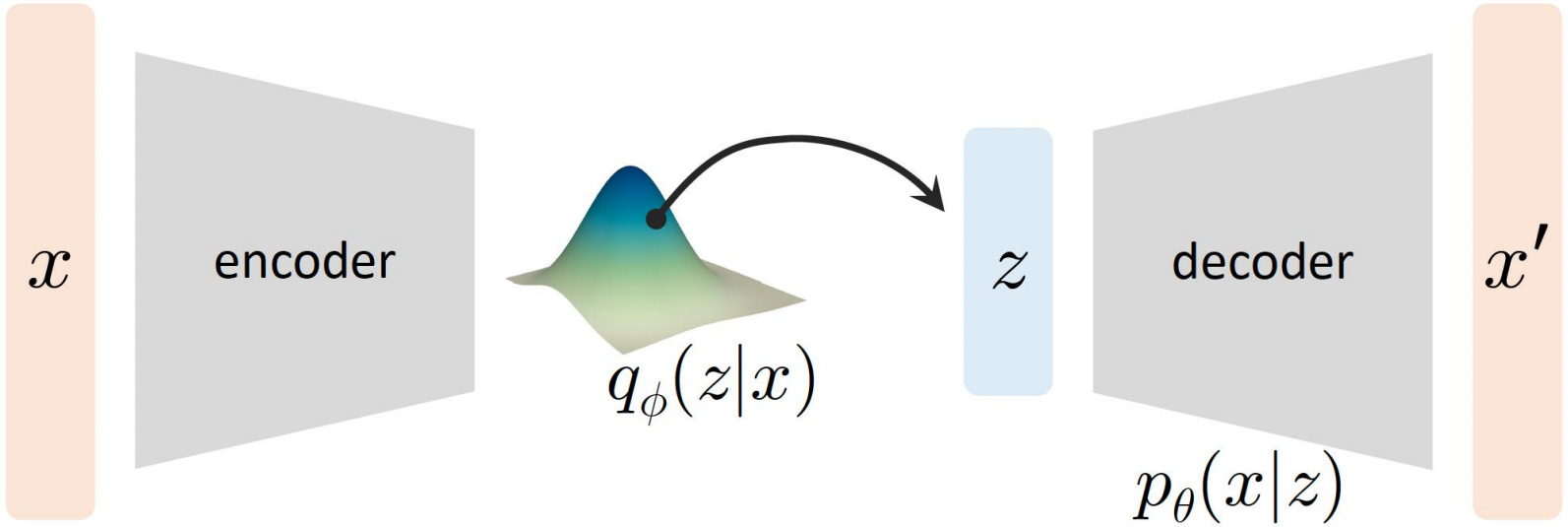
➢ **Question:** How to implement the objective function?

✓ Reconstruction Loss (重建损失): $\mathbb{E}_{q_\phi(z|x)}\left[\log\left(p_\theta(x \mid z)\right)\right]$

✓ Regularization Loss (正则损失): $D_{KL}\left(q_\phi(z \mid x) \parallel p(z)\right)$

Variational Autoencoder

- **Implementation: Objective Functions**



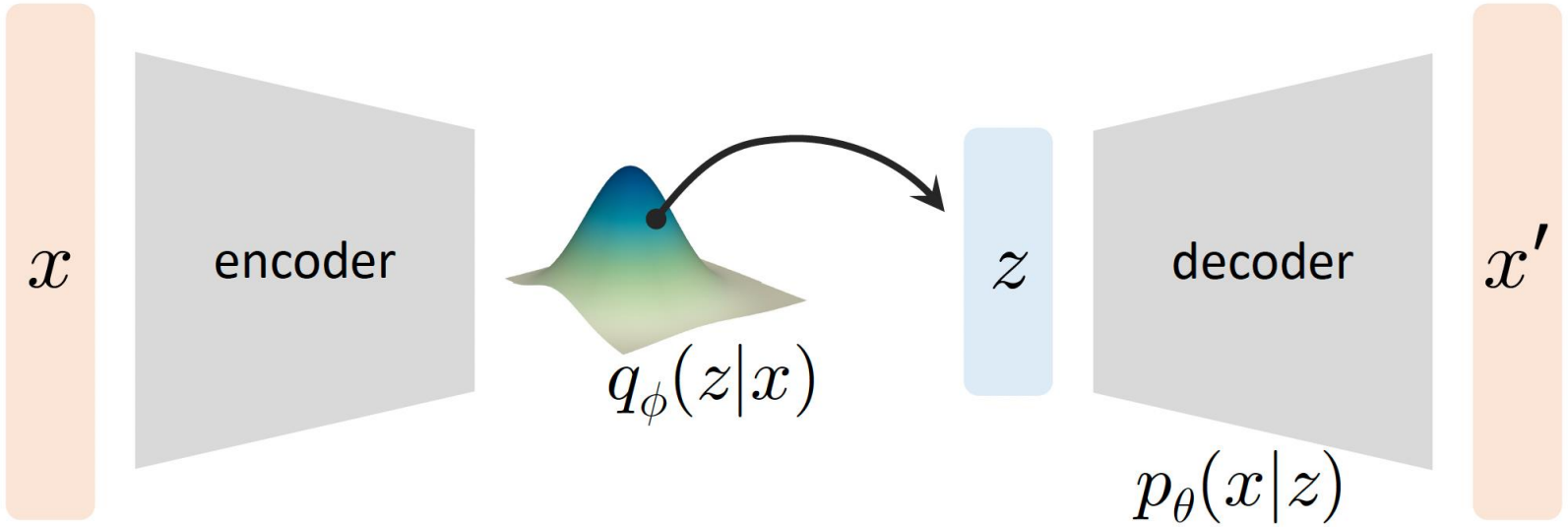
ELBO (θ, ϕ) $\quad \mathcal{L}(x;\theta,\phi) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - D_{KL}\left(q_\phi(z \mid x) \parallel p(z)\right)$

- **"Reconstruction Loss":** Maximizing log-likelihood

$$p_\theta(x \mid z) = \mathcal{N}(x'(z), \sigma_0^2 I) = \frac{1}{(2\pi\sigma_0^2)^{d/2}} \exp\left(-\frac{\|x - x'\|_2^2}{2\sigma_0^2}\right)$$

Variational Autoencoder

- **Implementation: Objective Functions**



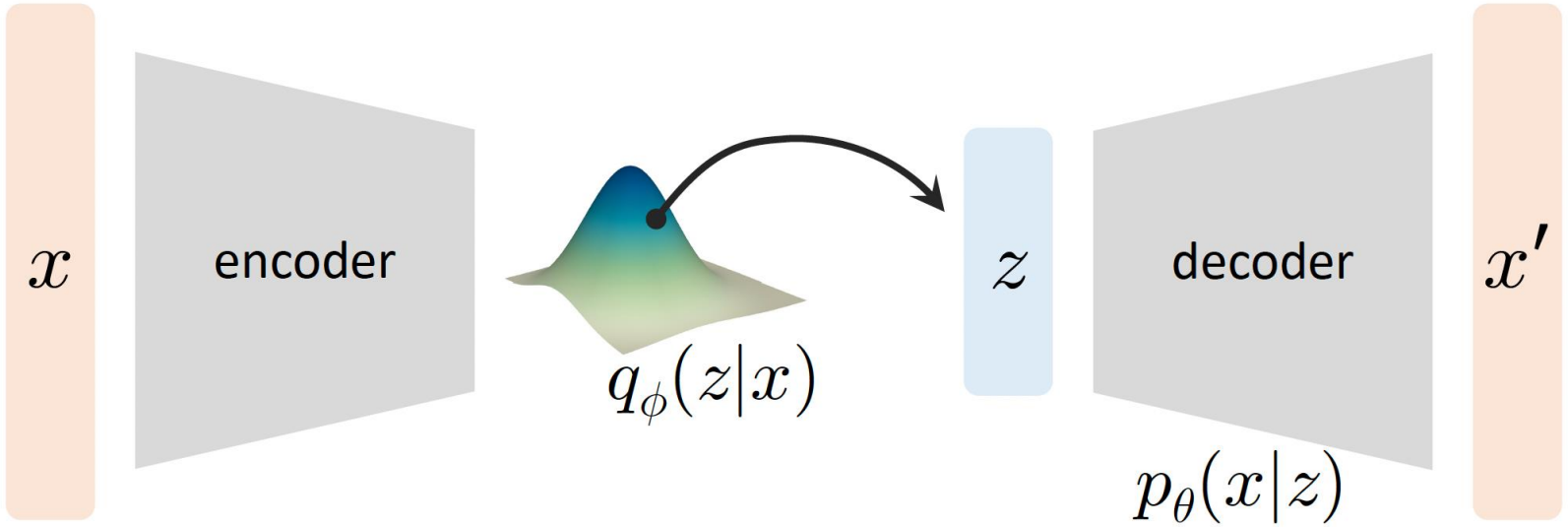
ELBO (θ, ϕ) $\quad \mathcal{L}(x;\theta,\phi) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - D_{KL}\left(q_\phi(z \mid x) \parallel p(z)\right)$

- **"Reconstruction Loss":** Maximizing log-likelihood ➔ L2 Loss

$$p_\theta(x \mid z) = \frac{1}{(2\pi\sigma_0^2)^{d/2}} \exp\left(-\frac{\|x - x'\|_2^2}{2\sigma_0^2}\right) \Rightarrow \log p_\theta(x \mid z) = -\frac{\|x - x'\|_2^2}{2\sigma_0^2} + \text{const}$$

Variational Autoencoder

- **Implementation: Objective Functions**

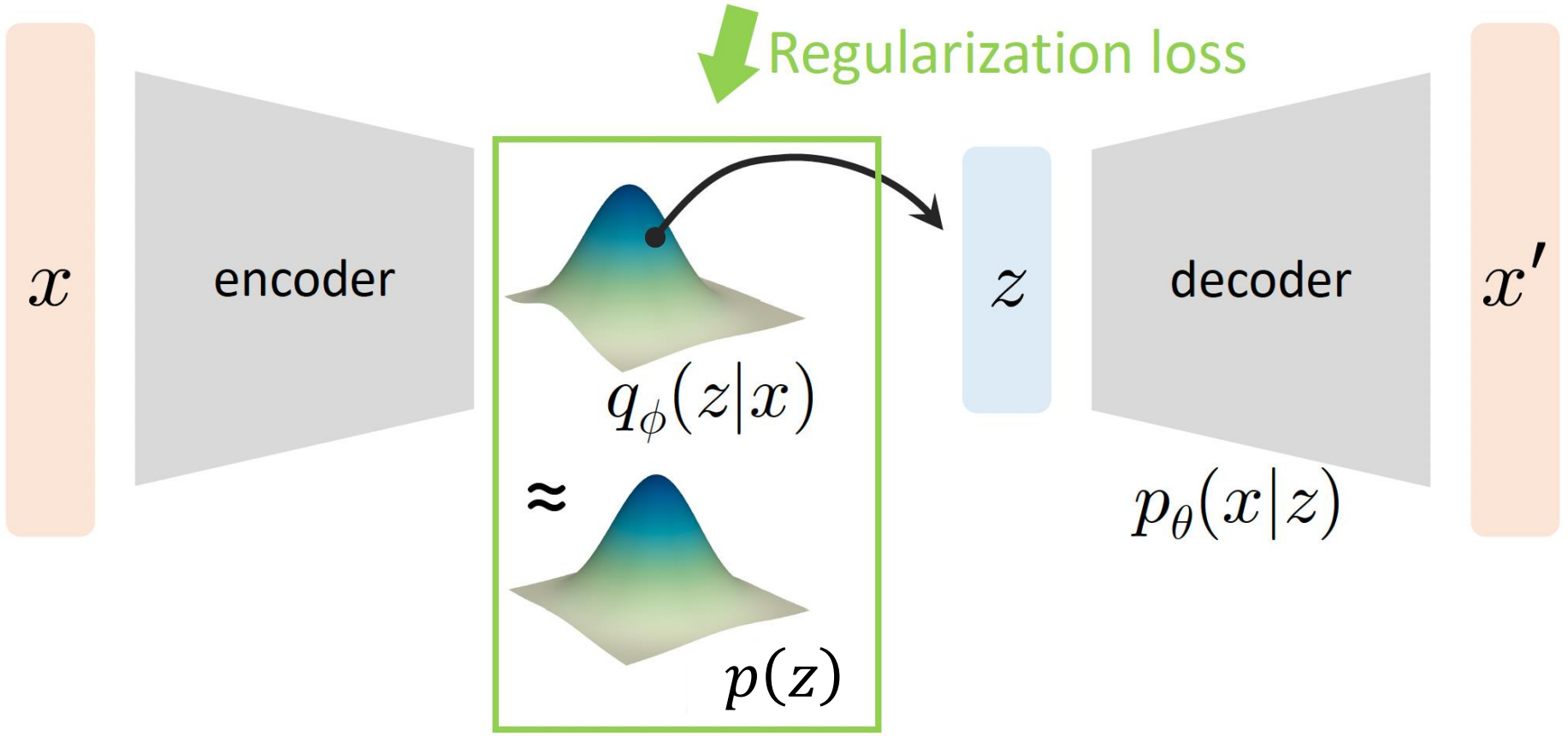


ELBO (θ, ϕ) $\quad \mathcal{L}(x;\theta,\phi) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - D_{KL}\left(q_\phi(z \mid x) \parallel p(z)\right)$

- **"Regularization Loss"**
  - ✓ Let $p(z)$ be an Gaussian prior $p(z) = \mathcal{N}(0, I)$
  - ✓ If $q_\phi(z \mid x)$ is Gaussian, $D_{KL}\left(q_\phi(z \mid x) \parallel p(z)\right)$ can be easily computed

Variational Autoencoder

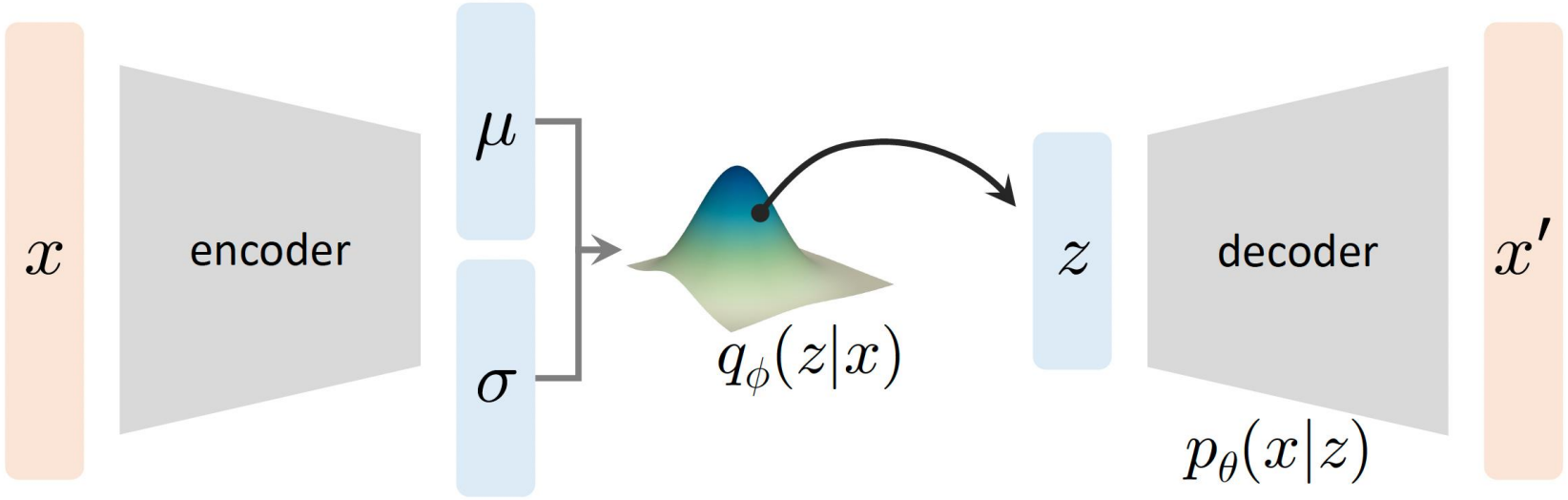
- **Implementation: Objective Functions**



ELBO (θ, ϕ) $\mathcal{L}(x;\theta,\phi) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - D_{KL}\left(q_\phi(z \mid x) \parallel p(z)\right)$

Variational Autoencoder

- **Training a Variational Autoencoder**



① Take a data point $x^i \in \mathcal{D}$

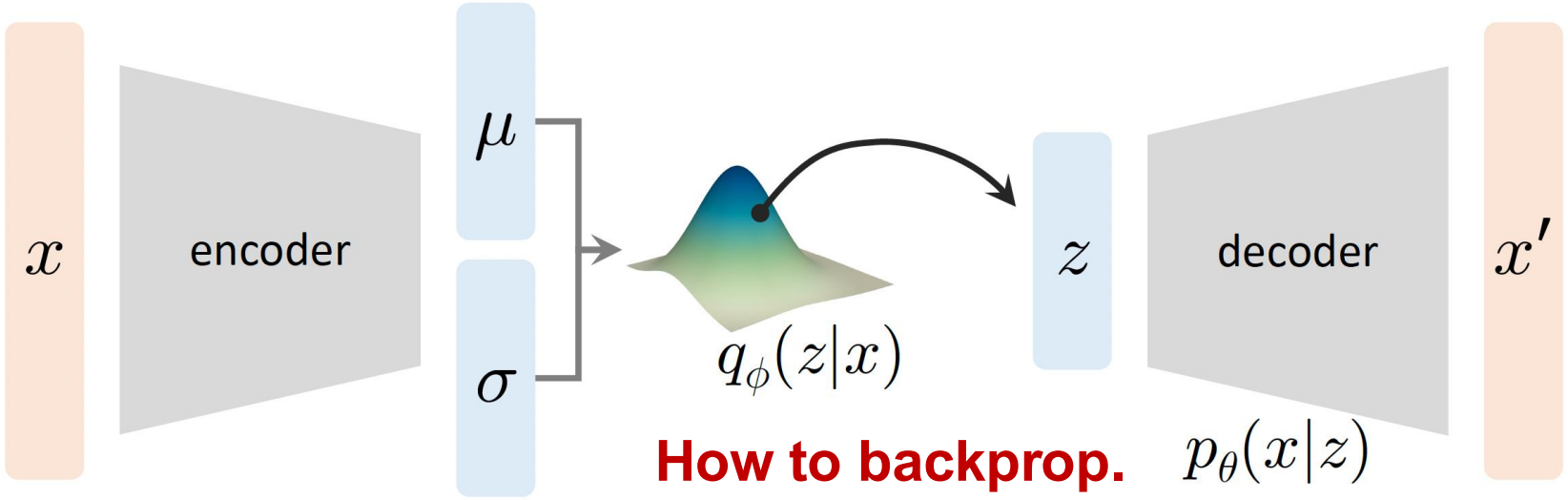
② Map it to $\hat{z}$ by sampling from the encoder $q_\phi(z \mid x^i)$

③ Pass $\hat{z}$ through the decoder and evaluate $\log p_\theta(x^i \mid \hat{z})$

④ Compute $\mathcal{L}(x^i; \theta, \phi)$ and update θ, ϕ by gradient descent

Variational Autoencoder

- **Training a Variational Autoencoder**



How to backprop. w.r.t. sampling?

① Take a data point $x^i \in \mathcal{D}$

② Map it to $\hat{z}$ by **sampling** from the encoder $q_\phi(z \mid x^i)$

③ Pass $\hat{z}$ through the decoder and evaluate $\log p_\theta(x^i \mid \hat{z})$

④ Compute $\mathcal{L}(x^i; \theta, \phi)$ and update θ, ϕ by **gradient descent**

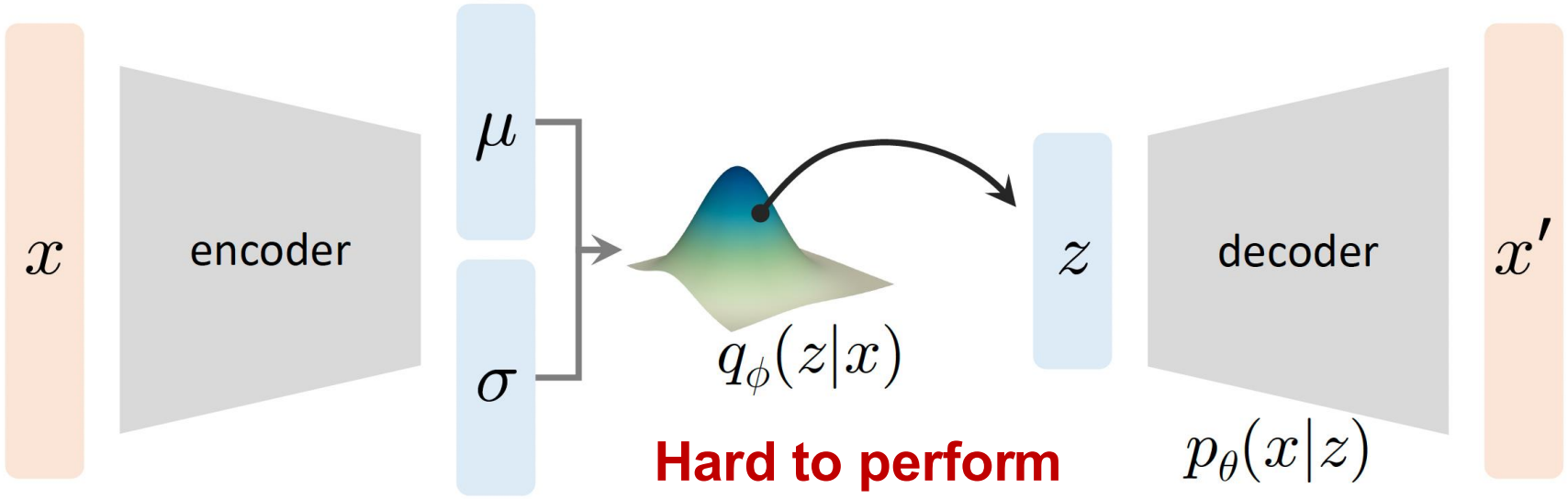


Lecture Outline

- Variational Learning
- Variational Autoencoder (VAE)
- Re-parameterization Trick
- The Generative Latent Space of VAE
- Advanced Variants of VAE

Re-parameterization Trick

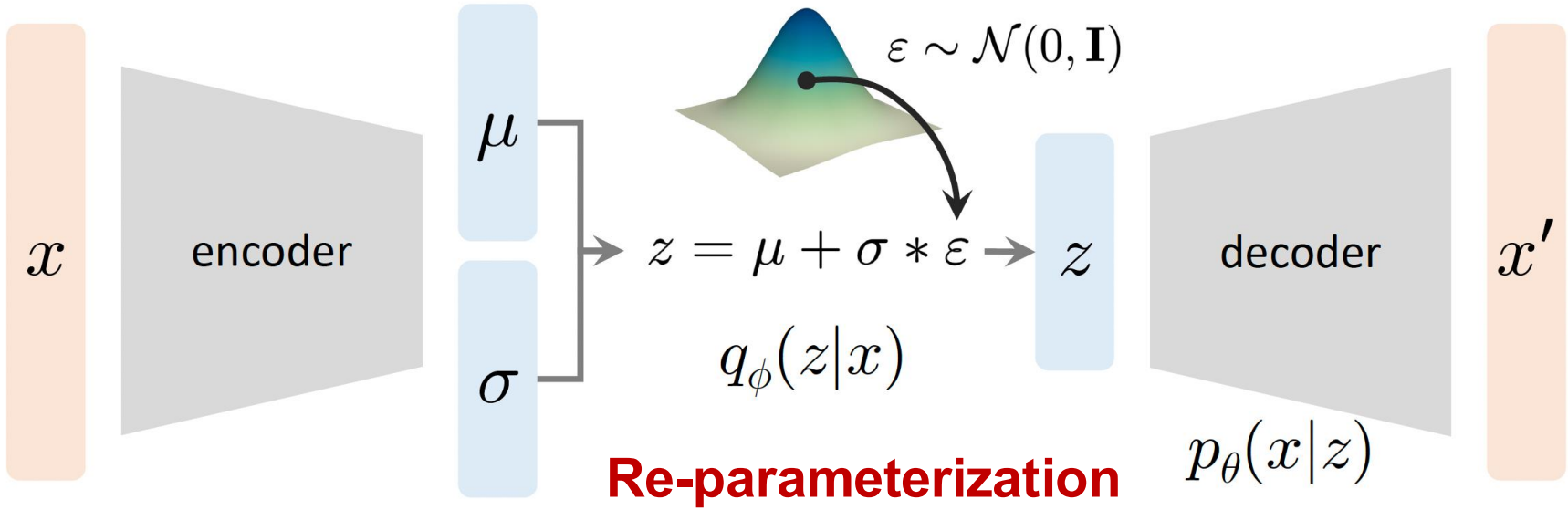
- **Re-parameterization Trick**
  - Given $q_\phi(z \mid x)$ is a Gaussian distribution $\mathcal{N}\left(z; \mu(x), \sigma(x)\right)$
    - ✓ Sample $z \sim q_\phi(z \mid x)$



Re-parameterization Trick

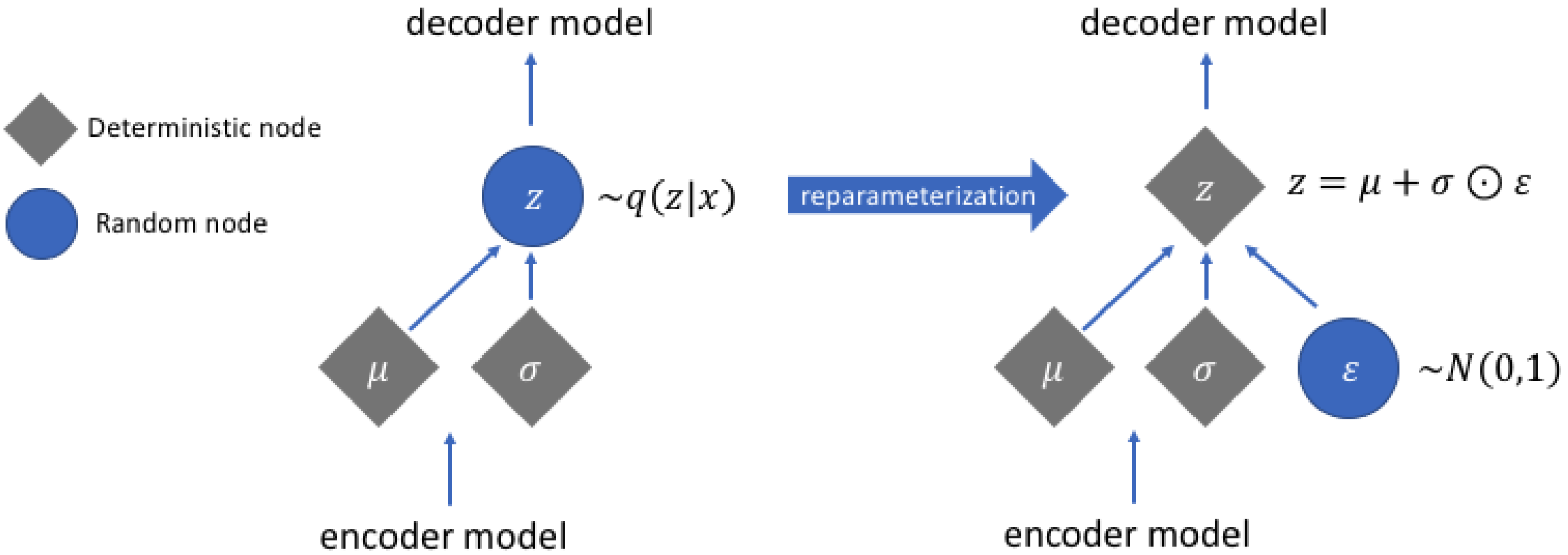
- **Re-parameterization Trick**
  - Given $q_\phi(z \mid x)$ is a Gaussian distribution $\mathcal{N}\left(z; \mu(x), \sigma(x)\right)$
    - ✓ Sample $z \sim q_\phi(z \mid x)$
    - ✓ Sample $\epsilon \sim \mathcal{N}(0, I)$, $z = \mu(x) + \sigma(x)\epsilon$

} Equivalent ways of sampling z



Re-parameterization Trick

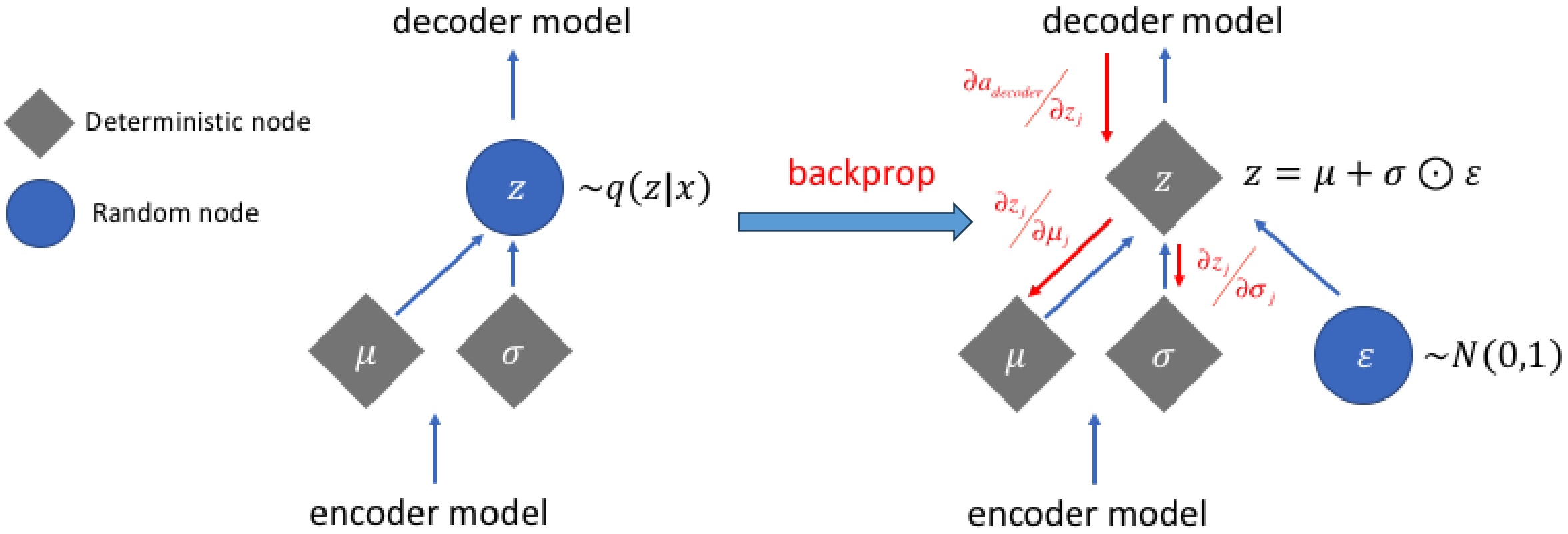
- **Re-parameterization Trick**



https://www.jeremyjordan.me/variational-autoencoders/

Re-parameterization Trick

- **Re-parameterization Trick**

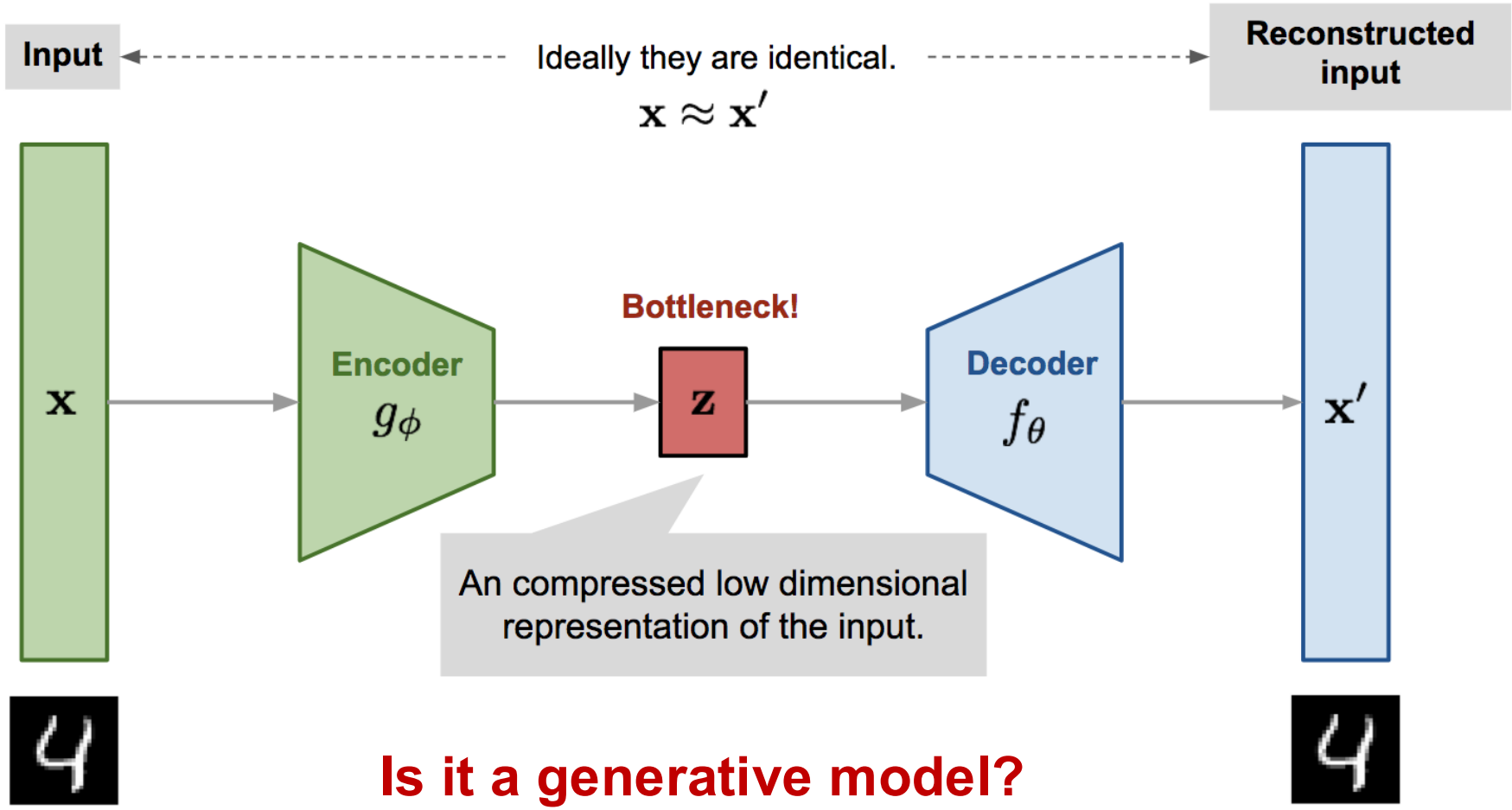


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The Generative Latent Space of VAE

- **Regularization-free Latent Space: Autoencoder (AE)**

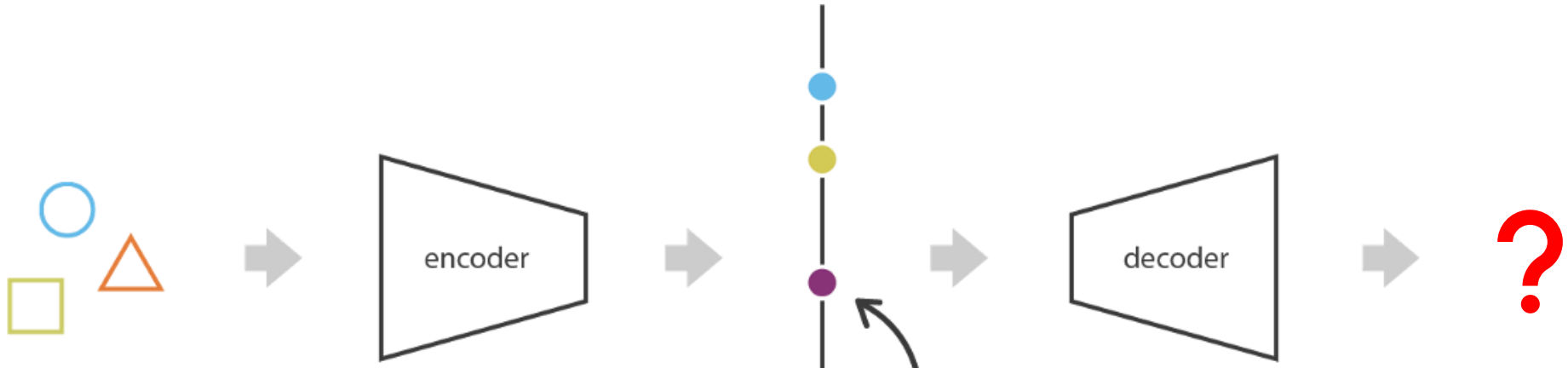


Is it a generative model?

https://lilianweng.github.io/posts/2018-08-12-vae/

The Generative Latent Space of VAE

- **Regularization-free Latent Space: Autoencoder (AE)**



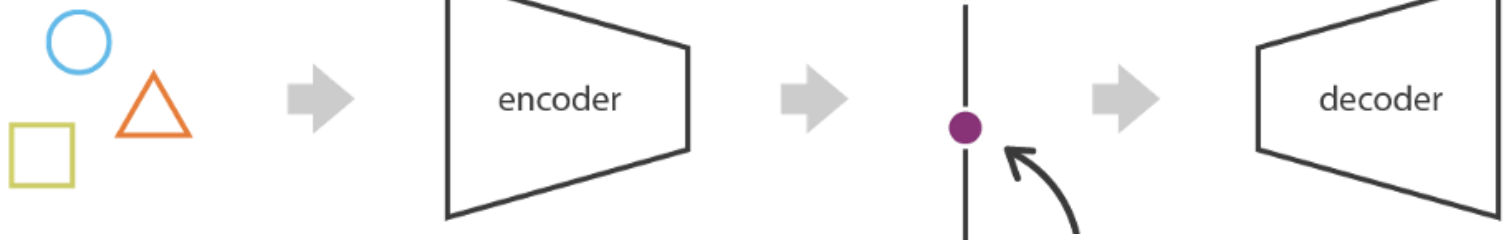
Training data for the Autoencoder

Codes sampled from the 1-D Latent Space

- https://towardsdatascience.com/understanding-variational-autoencoders-vaes-f70510919f73
- https://cloud.tencent.com/developer/article/1637439

The Generative Latent Space of VAE

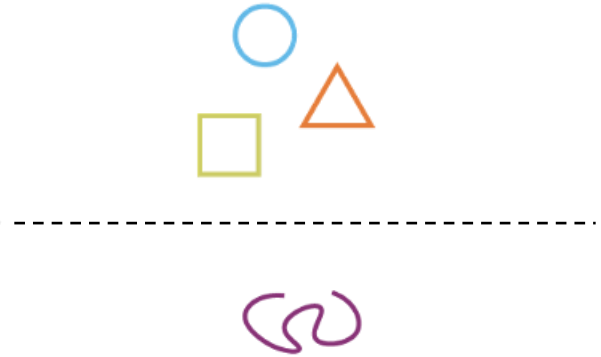
- **Regularization-free Latent Space: Autoencoder (AE)**



Training data for the Autoencoder

Codes sampled from the 1-D Latent Space

Training data can be reconstructed w/o loss

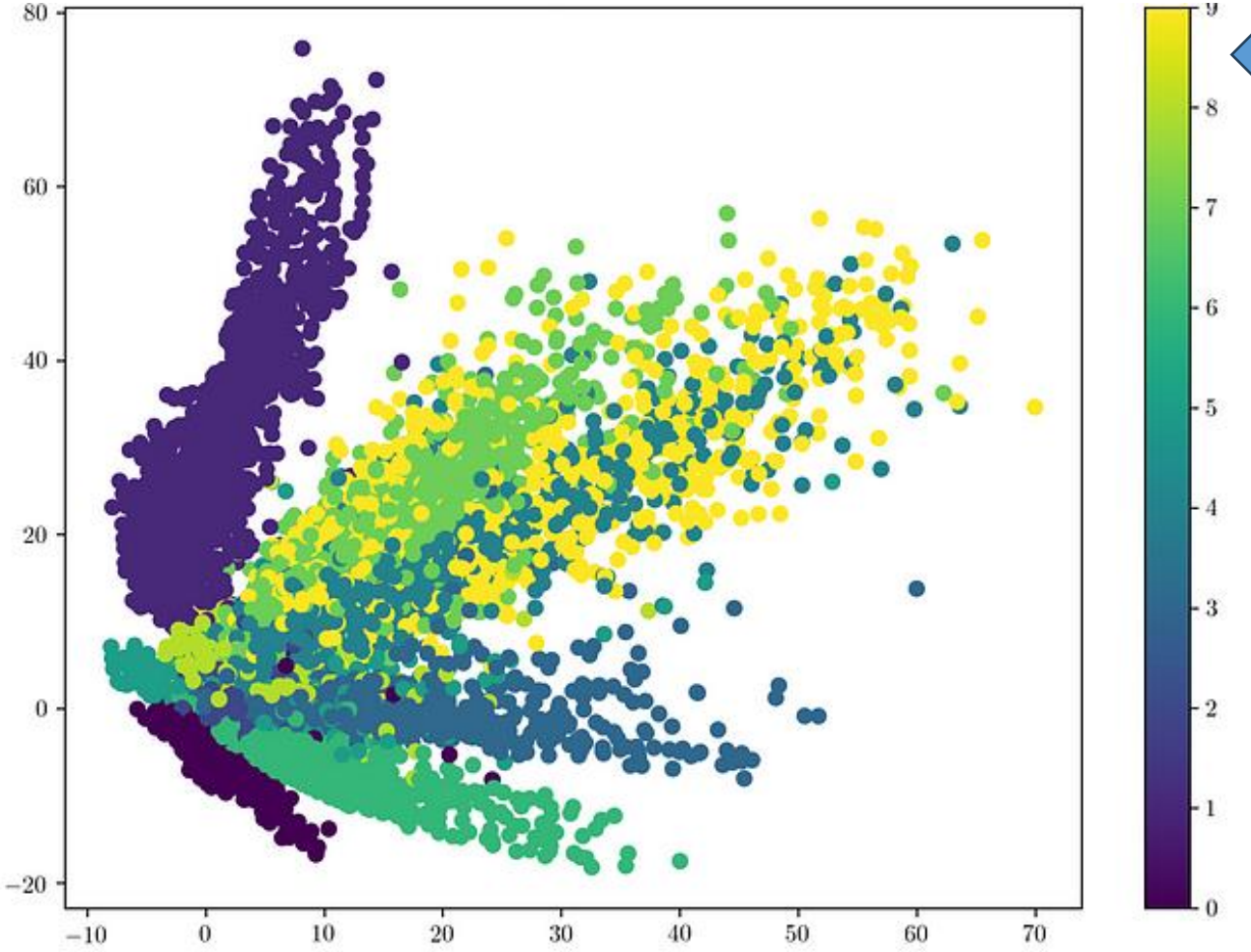


Reconstruction can be **meaningless** if not regularized

- https://towardsdatascience.com/understanding-variational-autoencoders-vaes-f70510919f73
- https://cloud.tencent.com/developer/article/1637439

The Generative Latent Space of VAE

- **Latent Space Analysis for AE**



Visualization of the Latent Space of AE on MNIST

① Latent codes are clustered based on their classes

② Some places are not properly encoded

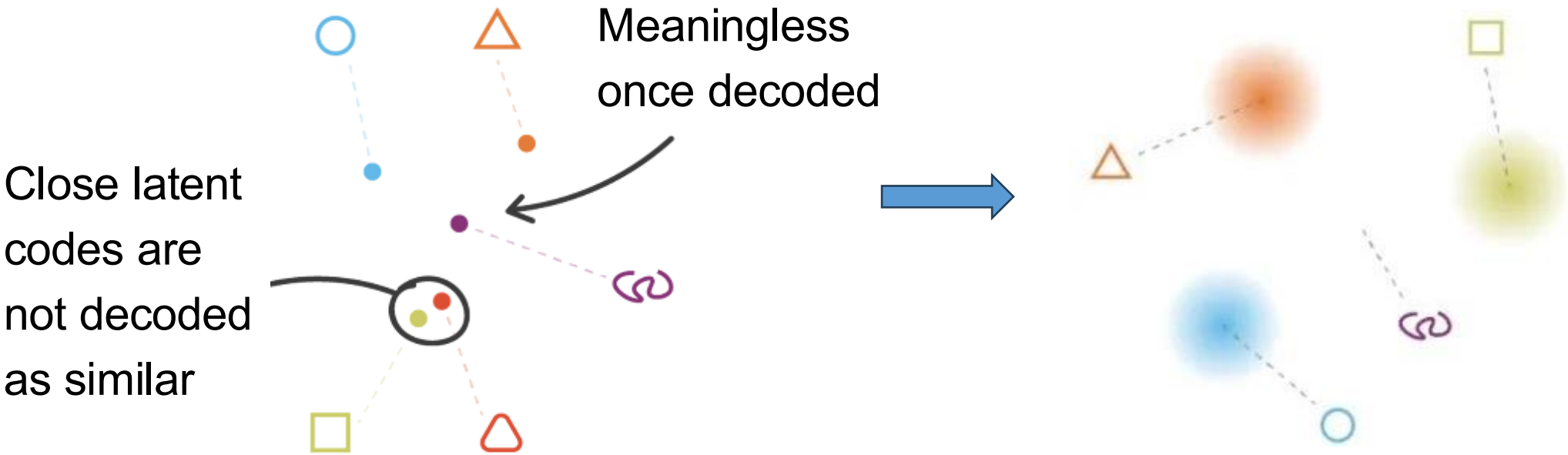
➢ **Question:**

How to fill the holes?

https://towardsdatascience.com/difference-between-autoencoder-ae-and-variational-autoencoder-vae-ed7be1c038f2/

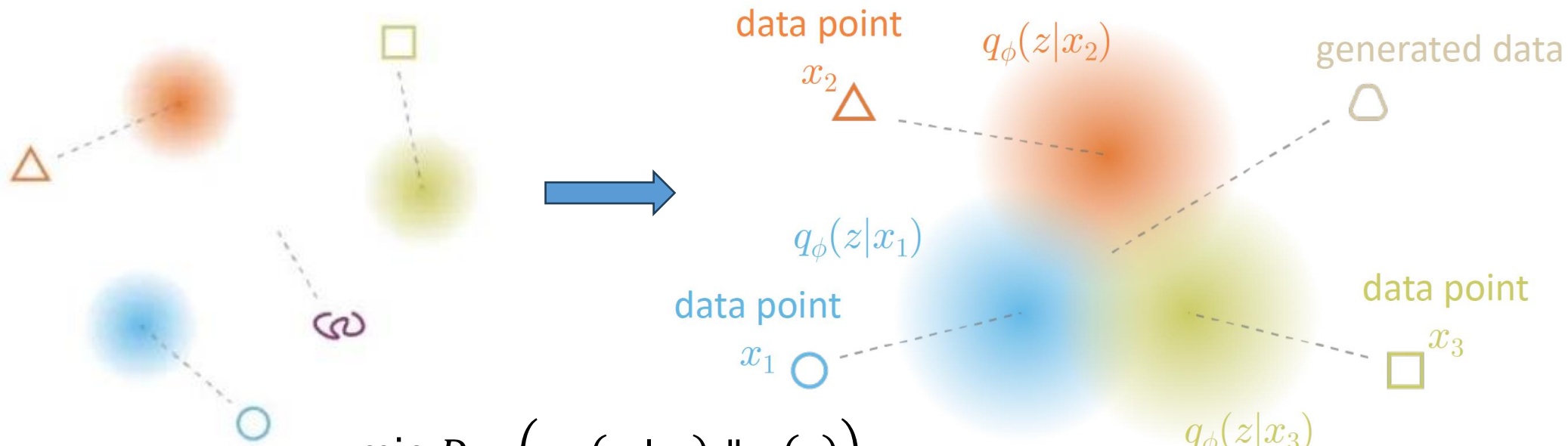
The Generative Latent Space of VAE

- **Latent Space Regularization**
  - ➢ **Challenge 1: Point-wise Encoding**
    - ✓ Decoder is trained only to reconstruct latent codes of training samples
    - ✓ Extend latent code $f_\phi(x)$ to **latent posterior distribution** $q_\phi(z \mid x)$



The Generative Latent Space of VAE

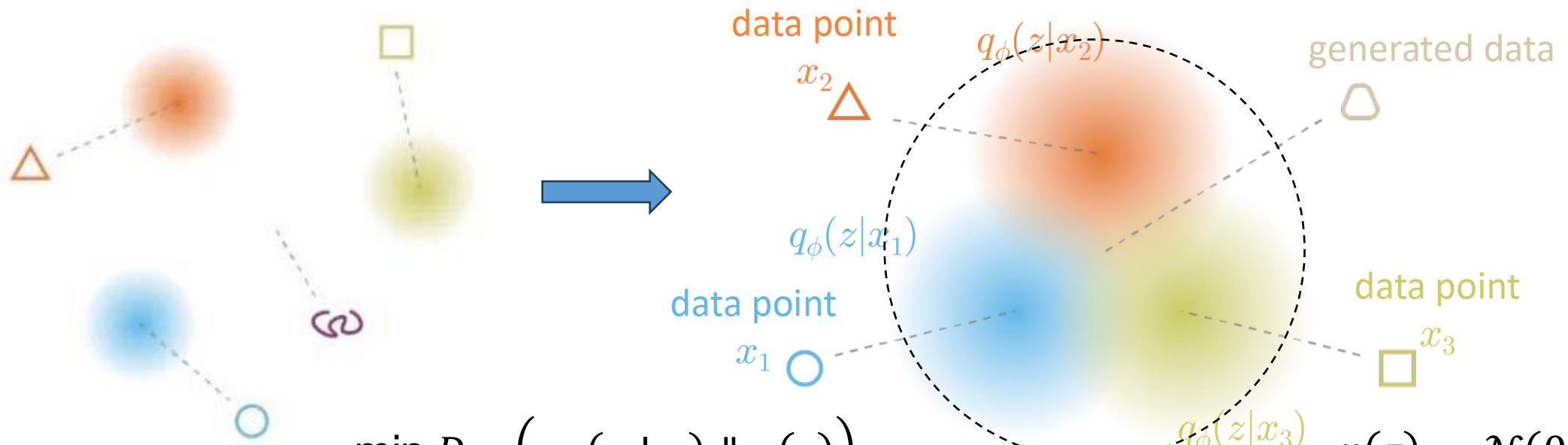
- **Latent Space Regularization**
  - ➢ **Challenge 2: Widely-separated Posterior Distributions**
    - ➔ The posterior distributions $q_\phi(z \mid x)$ may be widely-separated
    - ➔ Brings the posteriors closer by aligning them with the **shared prior $p(z)$**



$$\min D_{KL}\left(q_\phi(z \mid x) \parallel p(z)\right)$$

The Generative Latent Space of VAE

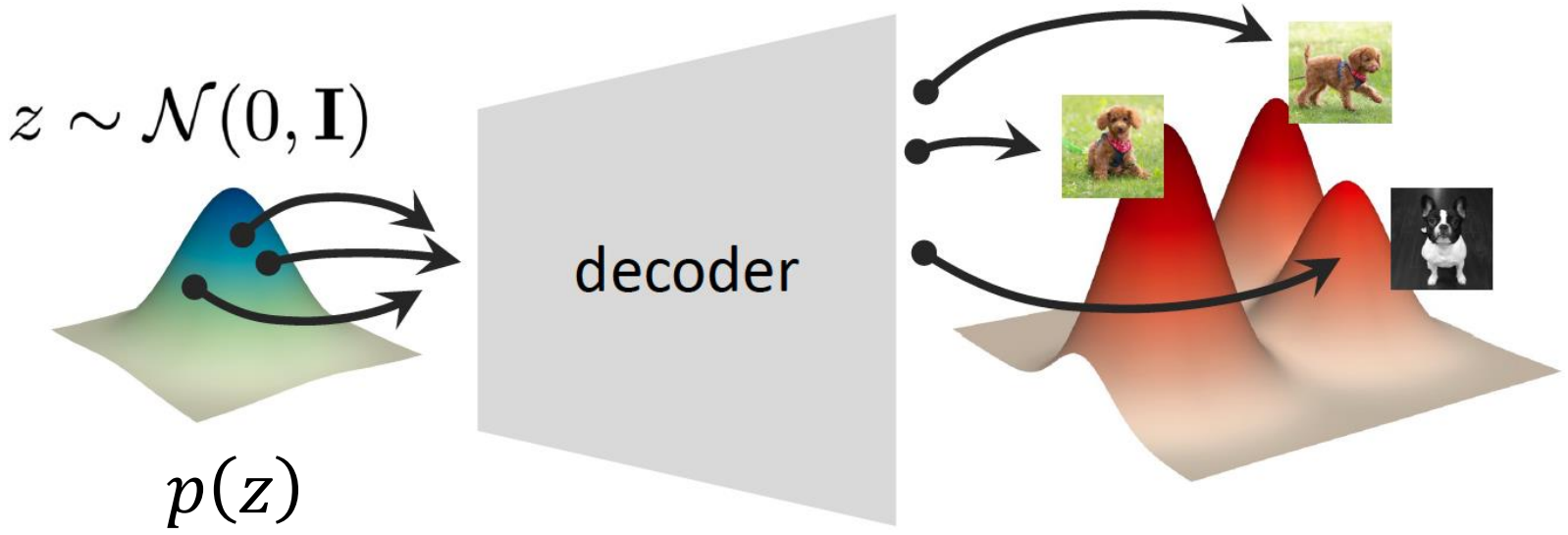
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    - ➔ Brings the posteriors closer by aligning them with the **shared prior $p(z)$**



$$\min D_{KL}\left(q_\phi(z \mid x) \parallel p(z)\right) \qquad p(z) = \mathcal{N}(0,1)$$

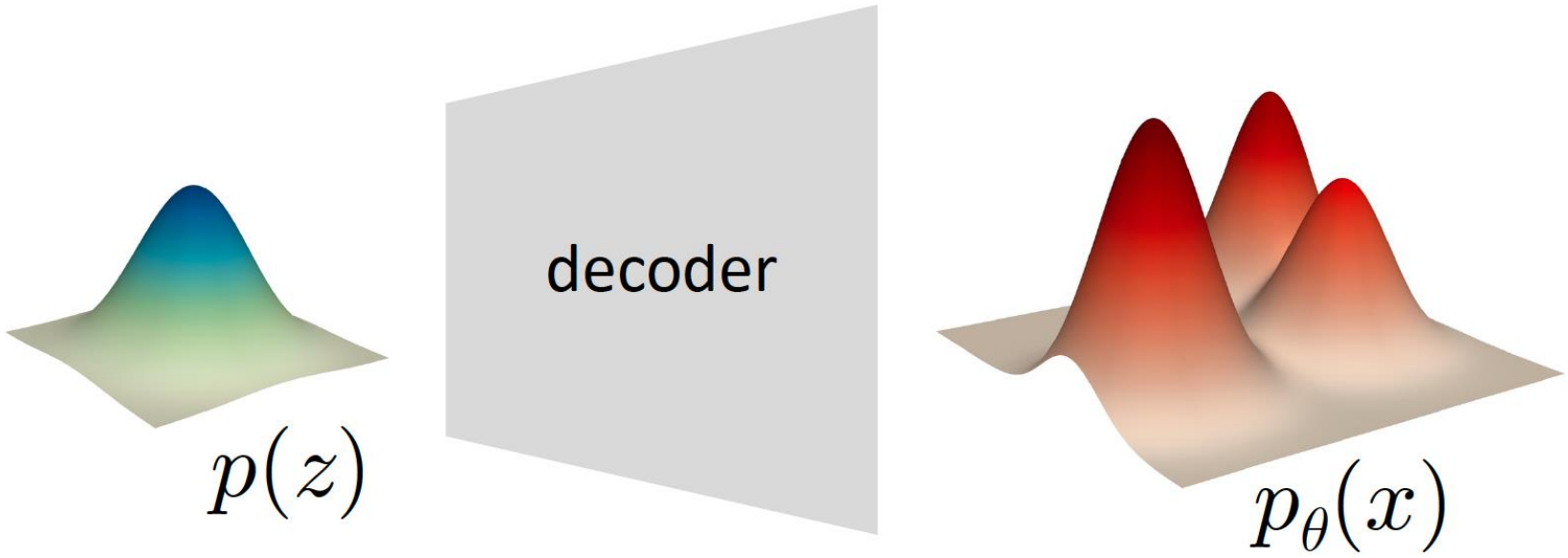
The Generative Latent Space of VAE

- **Latent Space Regularization**
  - **Challenge 3: Mismatch between Training and Generation**
    - ① Sample z from $p(z) = \mathcal{N}(0, I)$
    - ② Map z by the decoder net: $g_\theta(z)$



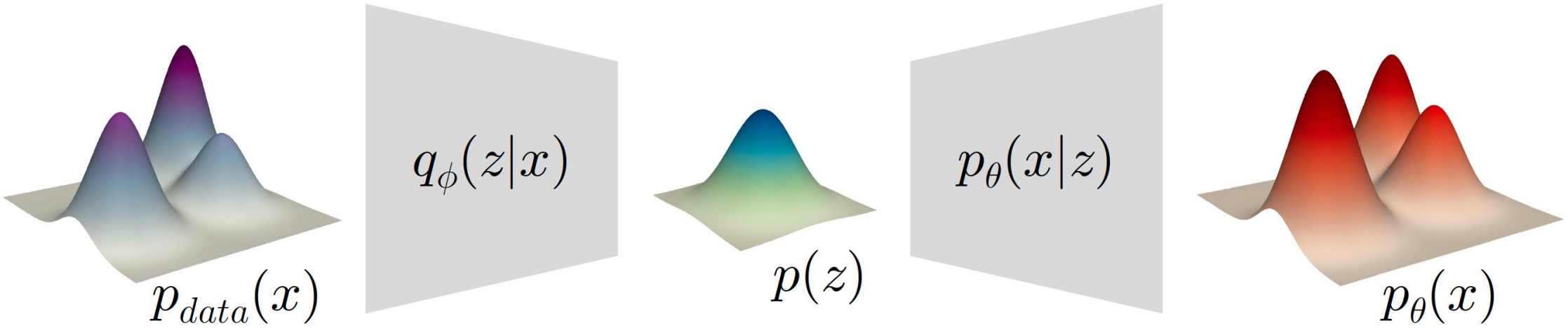
The Generative Latent Space of VAE

- **Latent Space Regularization**
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    - ✓ At generation time, the input distribution to decoder is $p(z)$



The Generative Latent Space of VAE

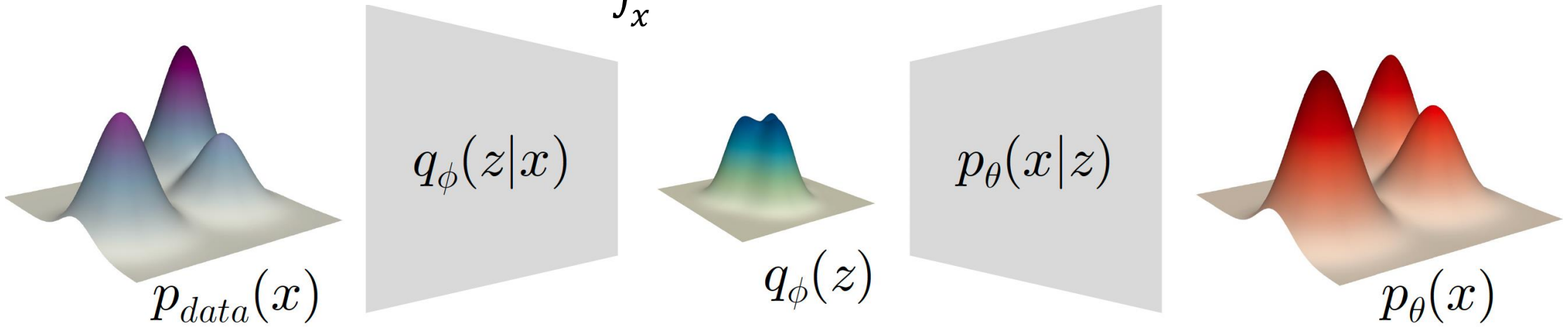
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  - ➢ **Challenge 3: Mismatch between Training and Generation**
    - ✓ At generation time, the input distribution to decoder is $p(z)$
    - ✓ **Question:** At train time, is the input distribution to decoder still $p(z)$?



The Generative Latent Space of VAE

- **Latent Space Regularization**
  - ➢ **Challenge 3: Mismatch between Training and Generation**
    - ✓ At generation time, the input distribution to decoder is $p(z)$
    - ✓ **Question:** At train time, is the input distribution to decoder still $p(z)$?
      - ➔ No! Needs to **regularize** the output distribution of encoder to $p(z)$

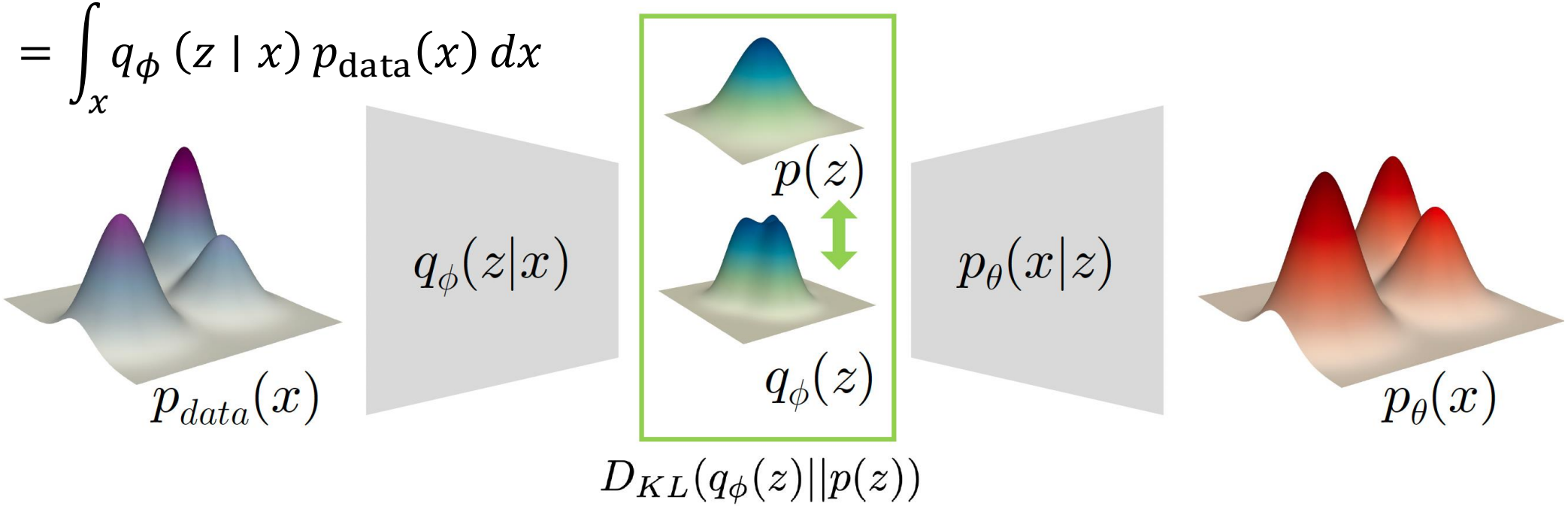
$$q_\phi(z) = \int_x q_\phi(z \mid x)\, p_{\text{data}}(x)\, dx$$



The Generative Latent Space of VAE

- **Latent Space Regularization**
  - ➢ **Challenge 3: Mismatch between Training and Generation**
    - ✓ At generation time, the input distribution to decoder is $p(z)$
    - ✓ Needs to **regularize** the output distribution of encoder to $p(z)$

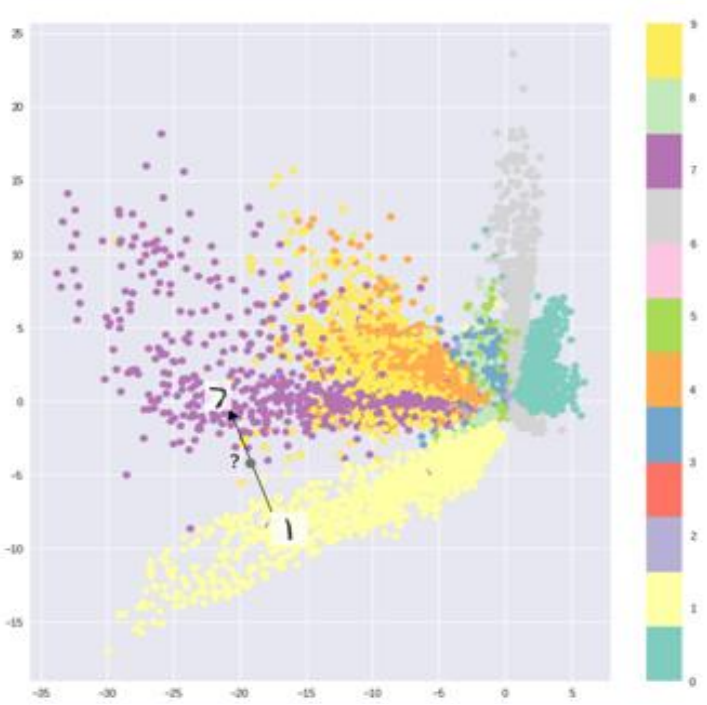
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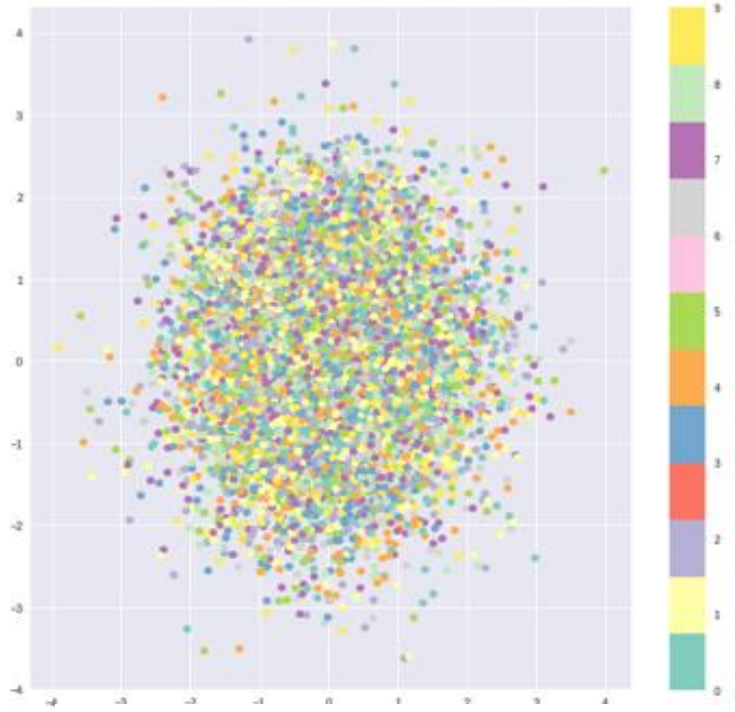
The Generative Latent Space of VAE

- **Latent Space Analysis for VAE**

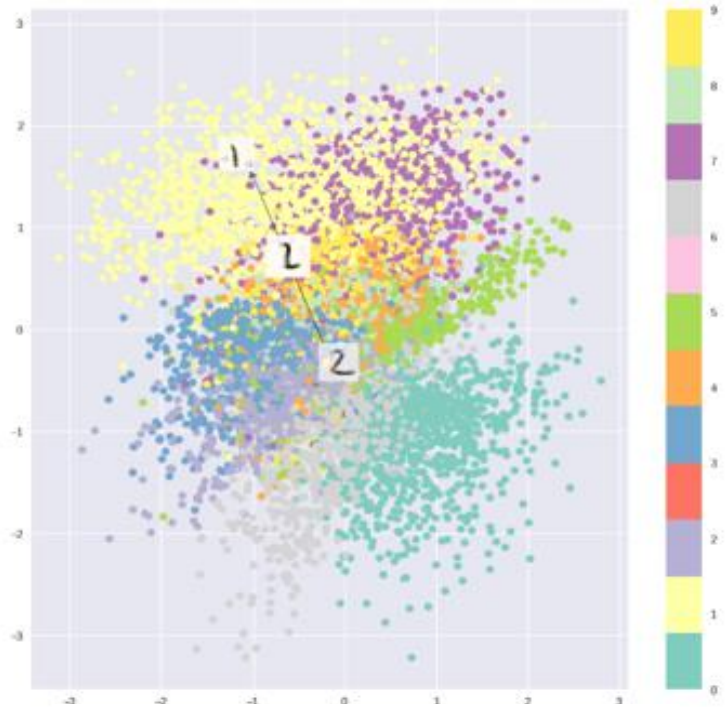
ELBO (θ, ϕ) $\mathcal{L}(x;\theta,\phi) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - D_{KL}\left(q_\phi(z \mid x) \parallel p(z)\right)$



Only with the Reconstruction Loss



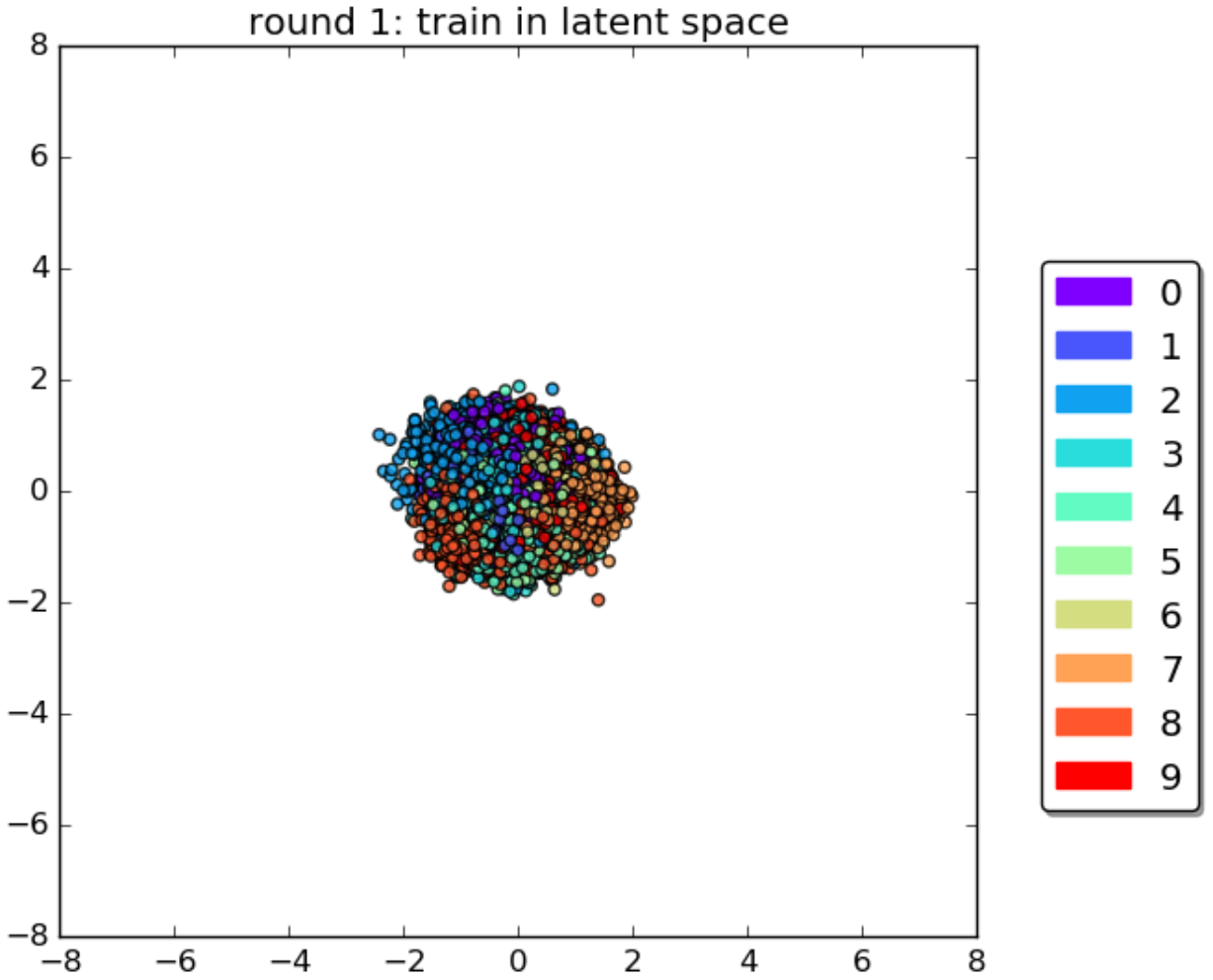
Only with the KL Regularization Loss



Reconstruction + KL Regularization

The Generative Latent Space of VAE

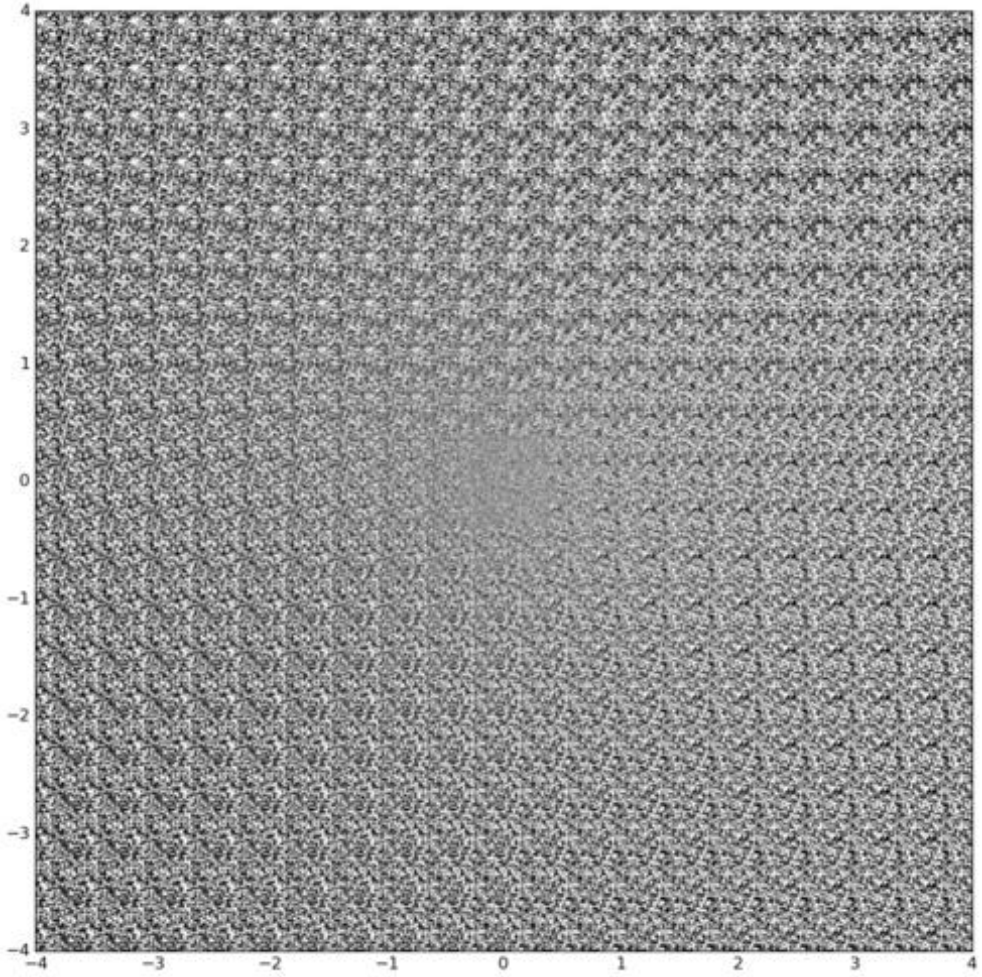
- **The Latent Space on MNIST: Encoder's Perspective**



https://blog.fastforwardlabs.com/2016/08/12/introducing-variational-autoencoders-in-prose-and-code.html

The Generative Latent Space of VAE

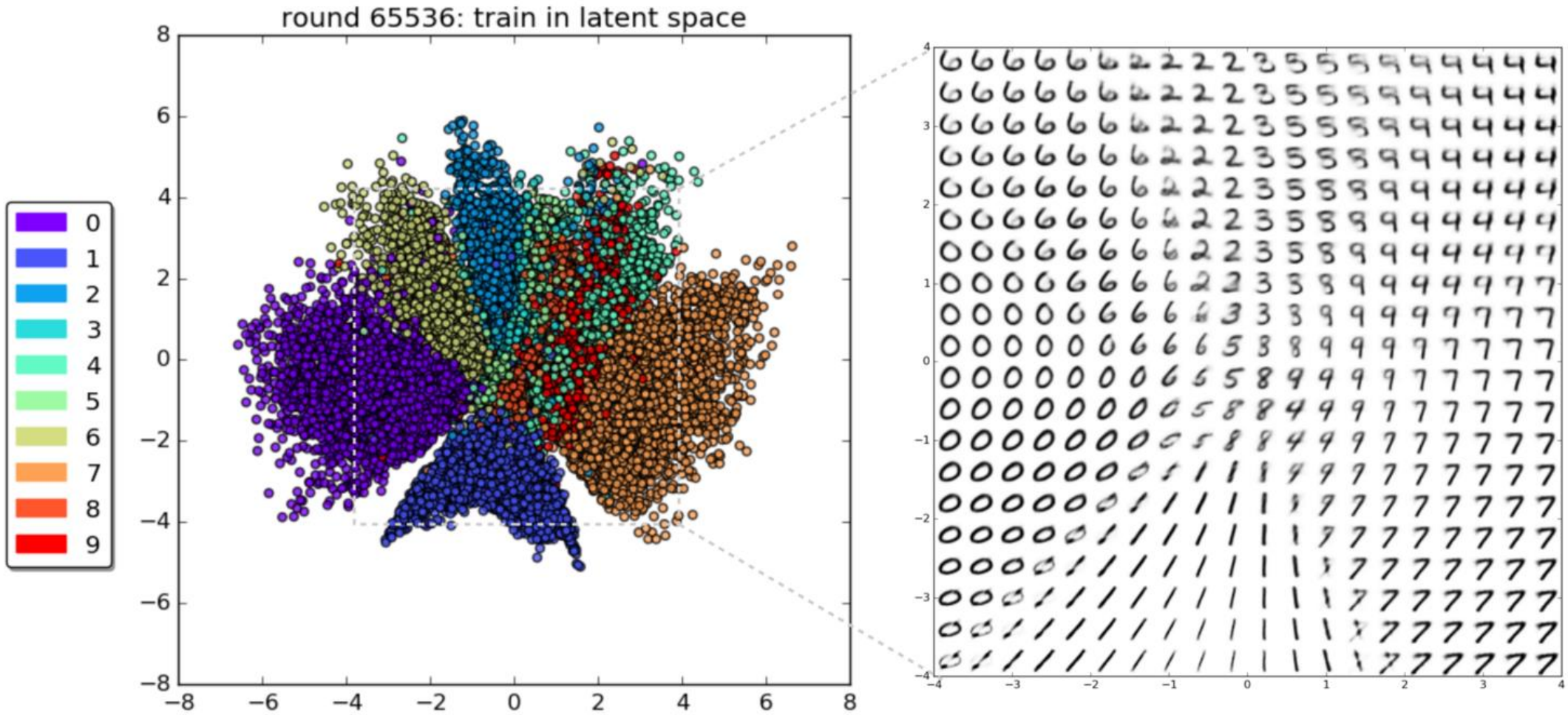
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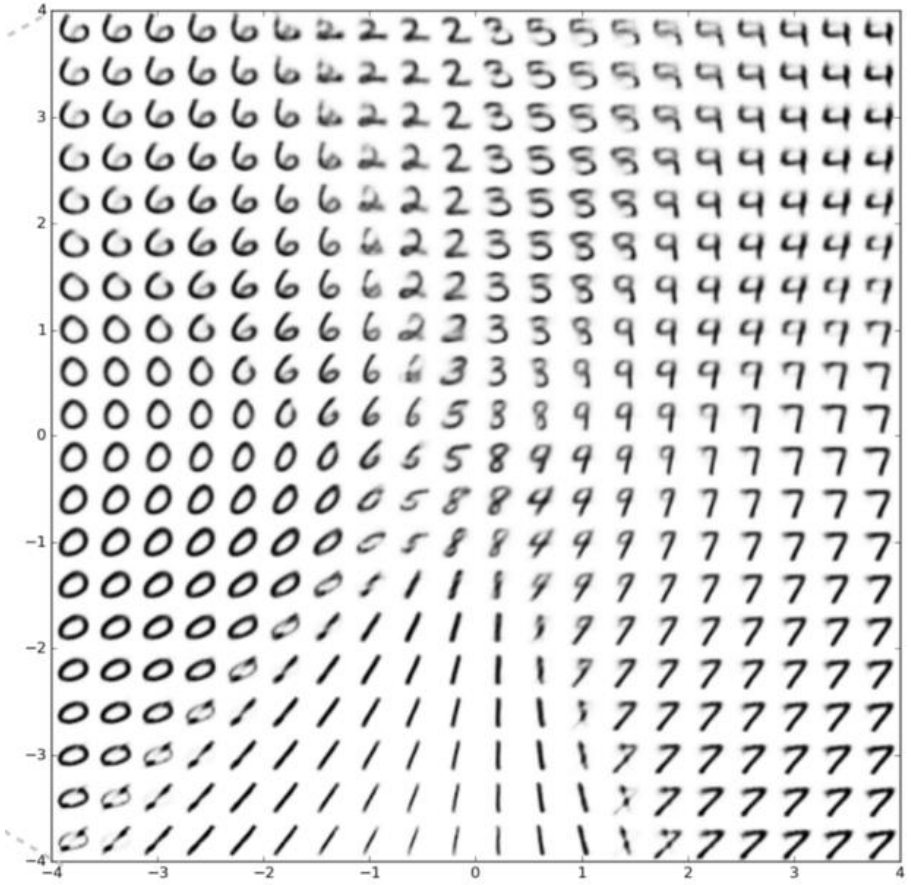
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https://blog.fastforwardlabs.com/2016/08/12/introducing-variational-autoencoders-in-prose-and-code.html

The Generative Latent Space of VAE

- **The Latent Space of VAE**



MNIST Dataset



Frey Face Dataset

The Generative Latent Space of VAE

- **Summary: VAE**
  - **Architecture**
    - ✓ Encoder $f_\phi(x) \to \big(\mu(x), \sigma(x)\big)$: $q_\phi(z \mid x) = \mathcal{N}\big(z; \mu(x), \sigma(x)\big)$
    - ✓ Decoder $g_\theta(z) \to x'(z)$: $p_\theta(x \mid z) = \mathcal{N}(x; x'(z), \sigma_0^2 I)$
  - **Objective Function:**

$$E_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p(\mathbf{x}|\mathbf{z};\theta)] - D_{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))$$

    - ✓ Reconstruction Loss: L2 loss if $p_\theta(x \mid z)$ is Gaussian
    - ✓ Regularization Loss: Align $q_\phi(z \mid x)$ with the prior $p(z)$
      - ① Gathered posterior distributions $q_\phi(z \mid x)$
      - ② Align the distribution of z in training and generation time



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- The Generative Latent Space of VAE
- Advanced Variants of VAE

Advanced Variants of VAE

- **Improvements of Vanilla VAE**
  - Learning Disentangled Latent Representation
  - Posterior Collapse
- **Learning Discrete Latent Code**
  - VQ-VAE
  - VQ-VAE2
- **Research Directions for Latent Variable Models**

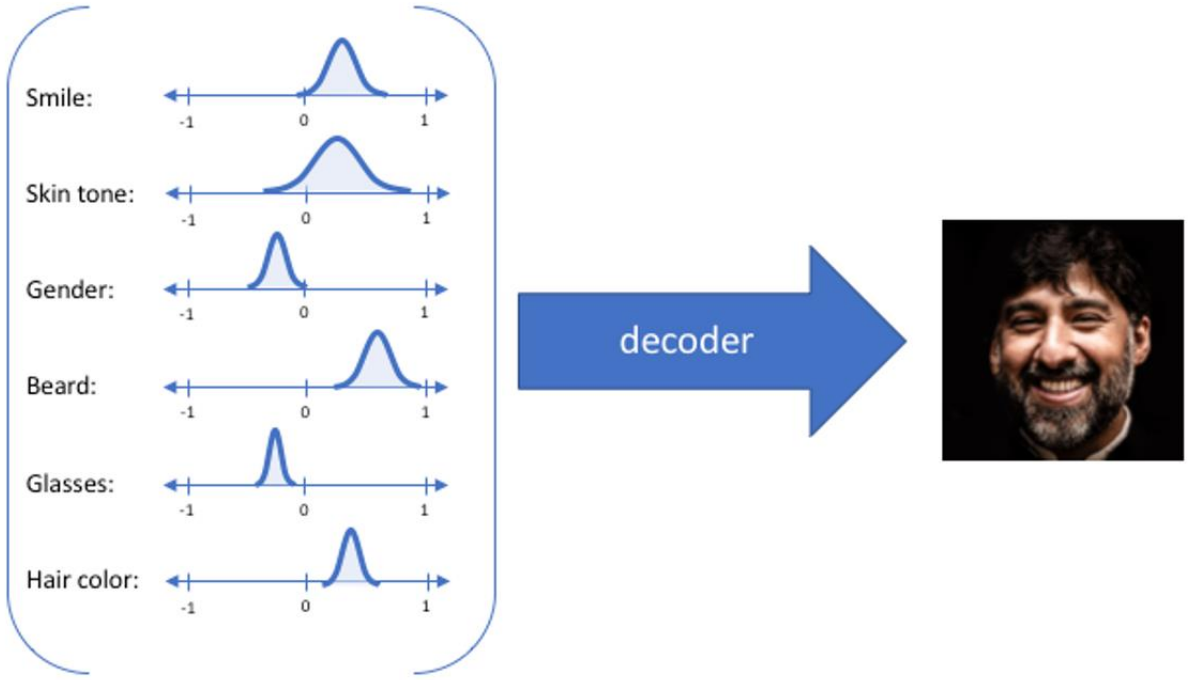
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Advanced Variants of VAE

- **Learning Disentangled Latent Representation**
  - **Disentanglement (解耦性)**

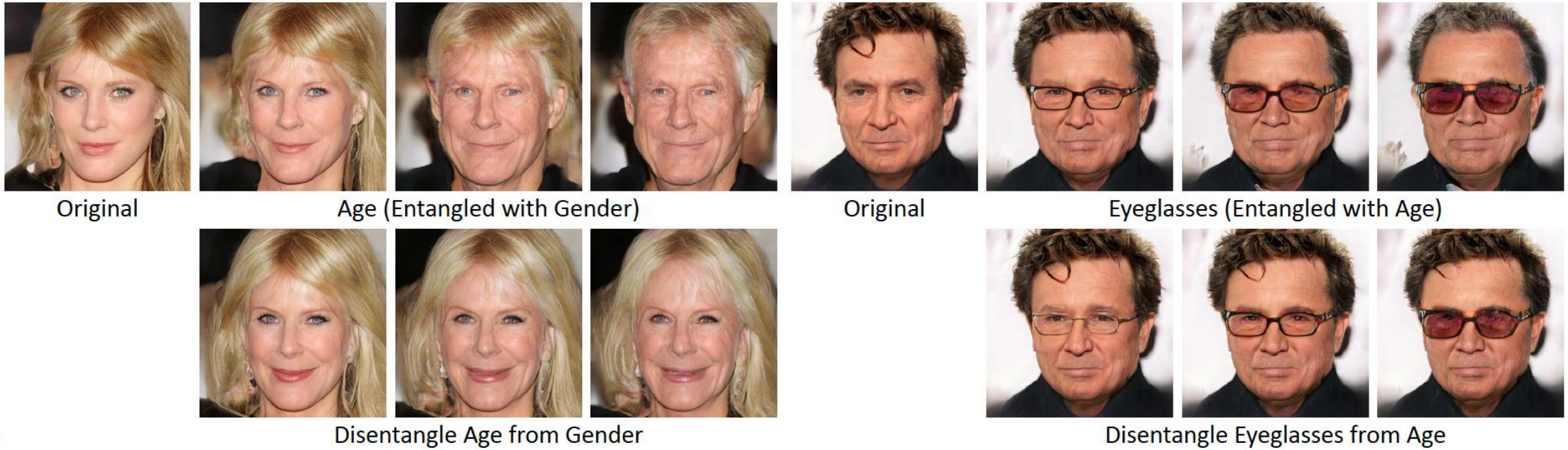
    Each latent dimension primarily captures **a single underlying factor of variation**, while remaining relatively invariant to changes in other factors



Advanced Variants of VAE

- **Learning Disentangled Latent Representation**
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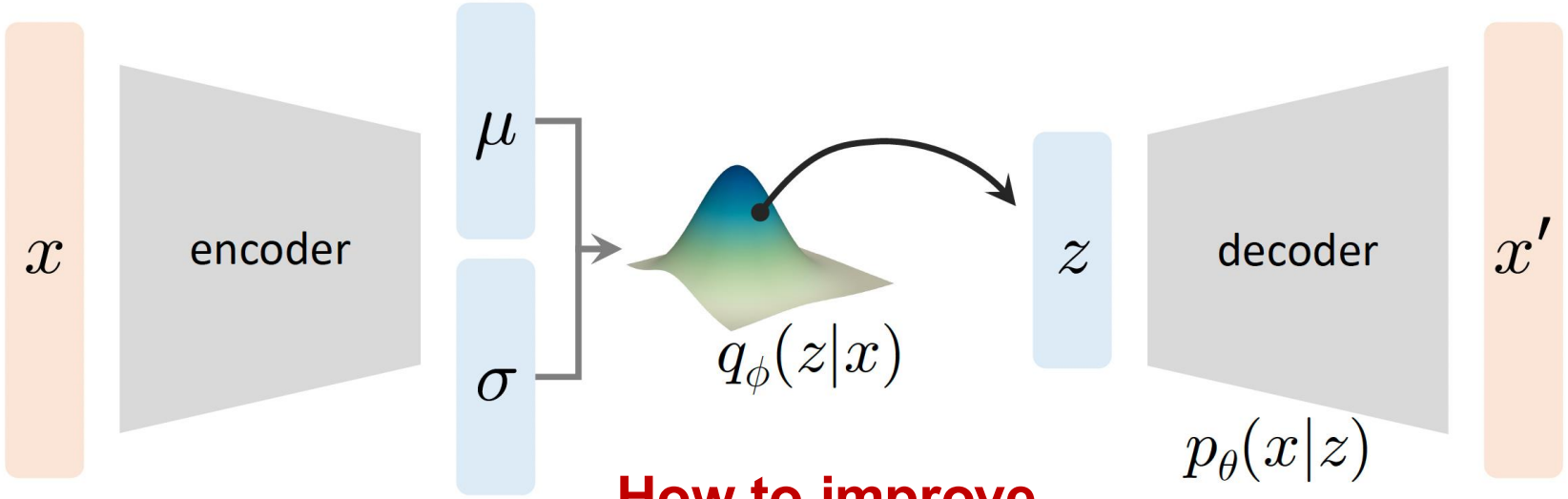


Yujun Shen et al., InterFaceGAN: Interpreting the Disentangled Face Representation Learned by GANs, TPAMI, 2021.

Advanced Variants of VAE

- **Learning Disentangled Latent Representation**
  - **Disentanglement (解耦性)**

    Each latent dimension primarily captures **a single underlying factor of variation**, while remaining relatively invariant to changes in other factors



How to improve disentanglement?

Advanced Variants of VAE

- **Learning Disentangled Latent Representation**
  - **Disentanglement (解耦性):** Each latent dimension primarily captures the single underlying factor of variation, while others remain unchanged

$$\max_{\phi,\theta} \mathbb{E}_{x\sim\mathbf{D}}\left[\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})]\right] \quad \text{subject to} \quad D_{KL}(q_\phi(\mathbf{z}|\mathbf{x})||p(\mathbf{z})) < \epsilon$$

Reconstruction Loss

Latent Information Bottleneck (ϵ: latent capacity)

① **Capacity Limit:** Forces z to contain the most principal generative factors

② **Independence Constratint:** $p(z) = \mathcal{N}(0, I)$ is a factorized independence target, which provides disentanglement pressure via KL loss

Irina Higgins et al., β-VAE: LEARNING BASIC VISUAL CONCEPTS WITH A CONSTRAINED VARIATIONAL FRAMEWORK, ICLR, 2017

Advanced Variants of VAE

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$$\max_{\phi,\theta} \mathbb{E}_{x\sim\mathbf{D}}\left[\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})]\right] \quad \text{subject to} \quad D_{KL}(q_\phi(\mathbf{z}|\mathbf{x})||p(\mathbf{z})) < \epsilon$$

$$\mathcal{F}(\theta,\phi,\beta;\mathbf{x},\mathbf{z}) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})] - \beta\left(D_{KL}(q_\phi(\mathbf{z}|\mathbf{x})||p(\mathbf{z})) - \epsilon\right)$$

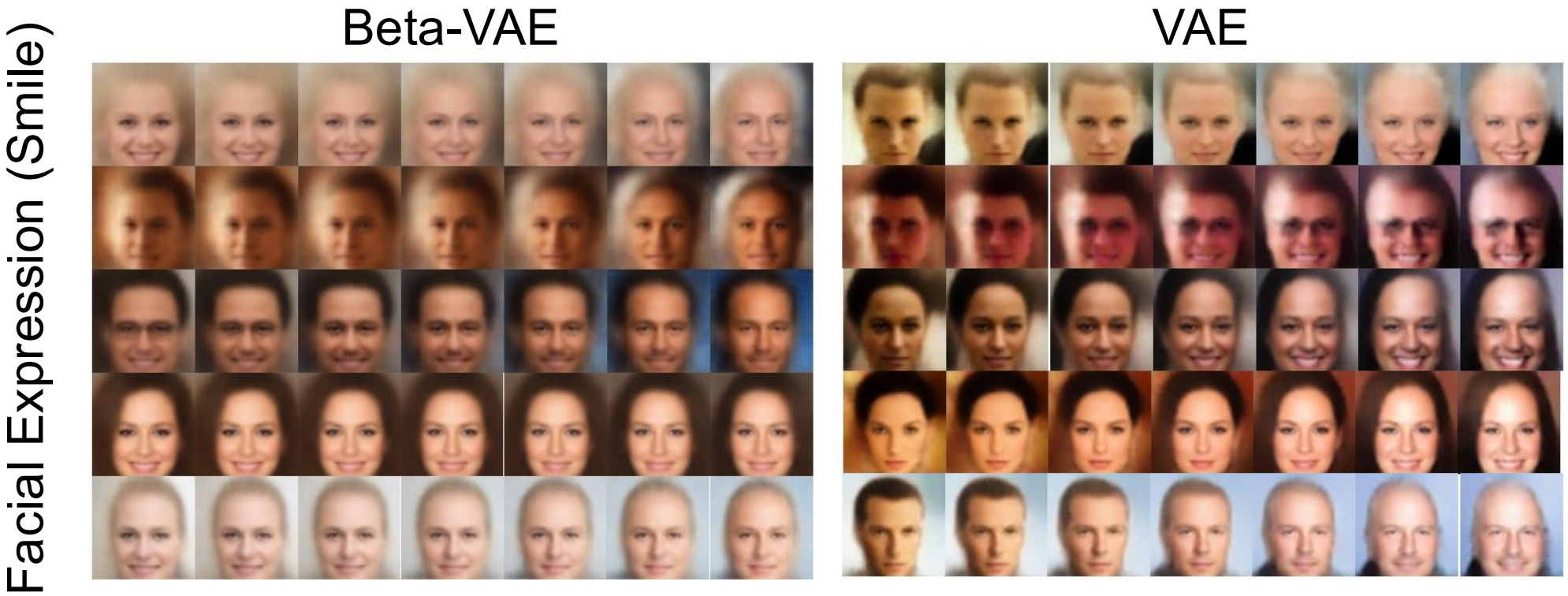
$$\mathcal{F}(\theta,\phi,\beta;\mathbf{x},\mathbf{z}) \geq \mathcal{L}(\theta,\phi;\mathbf{x},\mathbf{z},\beta) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})] - \beta\, D_{KL}(q_\phi(\mathbf{z}|\mathbf{x})||p(\mathbf{z}))$$

Weighted VAE Formulation

Irina Higgins et al., β-VAE: LEARNING BASIC VISUAL CONCEPTS WITH A CONSTRAINED VARIATIONAL FRAMEWORK, ICLR, 2017

Advanced Variants of VAE

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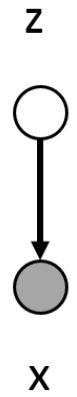
Irina Higgins et al., β-VAE: LEARNING BASIC VISUAL CONCEPTS WITH A CONSTRAINED VARIATIONAL FRAMEWORK, ICLR, 2017

Advanced Variants of VAE

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  - **Posterior Collapse**
- **Learning Discrete Latent Code**
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Advanced Variants of VAE

- **Posterior Collapse**
  - Formulation of Latent Variable Models



$$p(x, z) = p(x \mid z)\, p(z)$$

  - **Goal:** Help generative modeling by Introducing z

$$p(x) = \int p(z)\ p(x|z)\, dz$$

  - **Learning Objective:** Reconstruction + Regularization

$$\mathcal{L}(x; \theta, \phi) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - D_{KL}\left(q_\phi(z \mid x) \parallel p(z)\right)$$

Does it always work in practice?

Advanced Variants of VAE

- **Posterior Collapse**

$$\mathcal{L}(x;\theta,\phi) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - D_{KL}\left(q_\phi(z \mid x) \parallel p(z)\right)$$

① **Case1:** A weak decoder has not learned to use z effectively

✓ Reconstruction loss is large at current step

✓ KI regularization pushes the posterior $q_\phi(z \mid x)$ toward the prior $p(z)$

➔ Training may get stuck with uninformative z and poor reconstruction

Advanced Variants of VAE

- **Posterior Collapse**

$$\mathcal{L}(x;\theta,\phi) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - D_{KL}\left(q_\phi(z \mid x) \parallel p(z)\right)$$

① **Case1:** A weak decoder has not learned to use z effectively

✓ Reconstruction loss is large at current step

✓ KI regularization pushes the posterior $q_\phi(z \mid x)$ toward the prior $p(z)$

② **Case2:** A powerful autoregressive decoder can **bypass** z

✓ During training, it predicts each x_i using the true preceding context $x_{<i}$

$$p_\theta(x \mid z) = \prod_i p_\theta\left(x_i \mid x_{<i}, z\right)$$

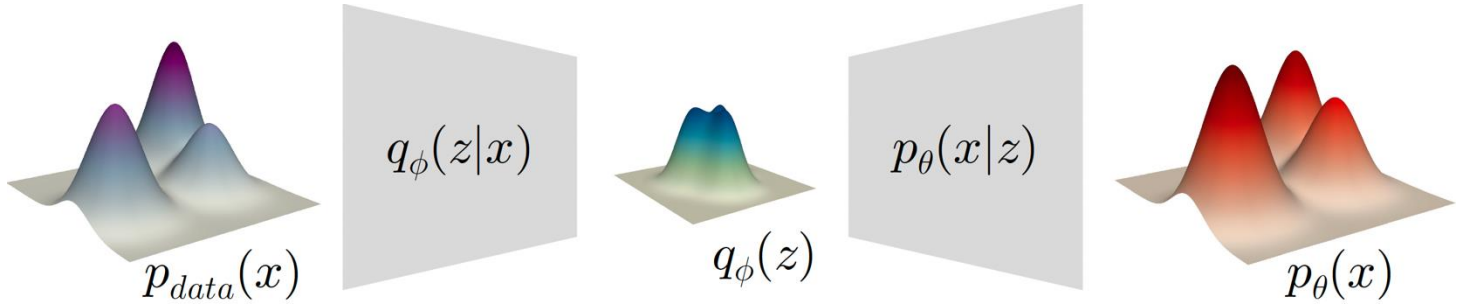
✓ A low reconstruction loss can be achieved while **ignoring** z

Advanced Variants of VAE

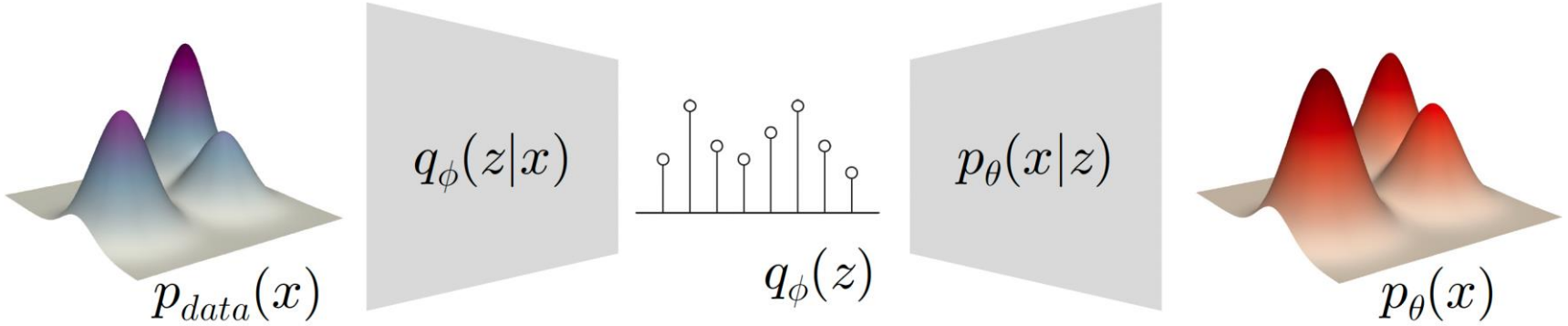
- **Improvements of Vanilla VAE**
  - Learning Disentangled Latent Representation
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Advanced Variants of VAE

- **Recap**
  - **Original VAE:** Latent variables are continuous

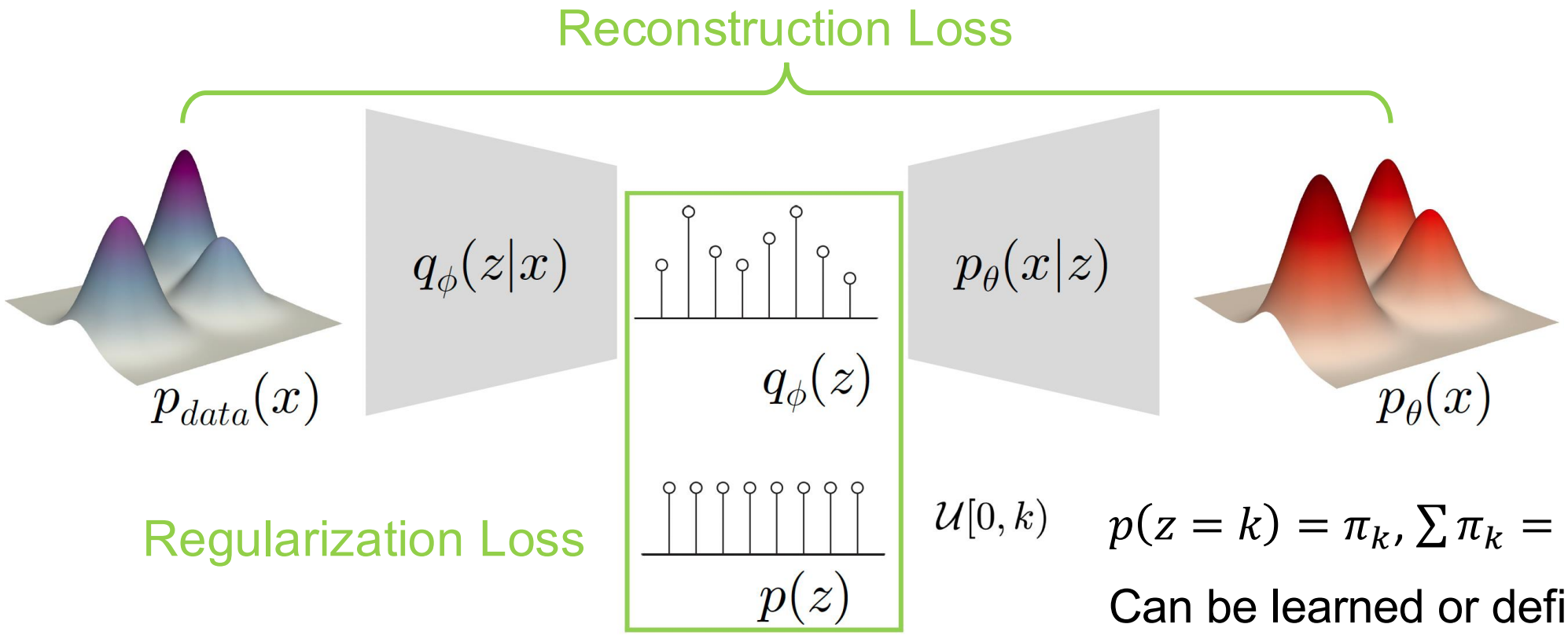


  - **Discrete VAE:** Use **discrete latent variables** to model broader data types
    - ✓ E.g.: language, speech, planning, …



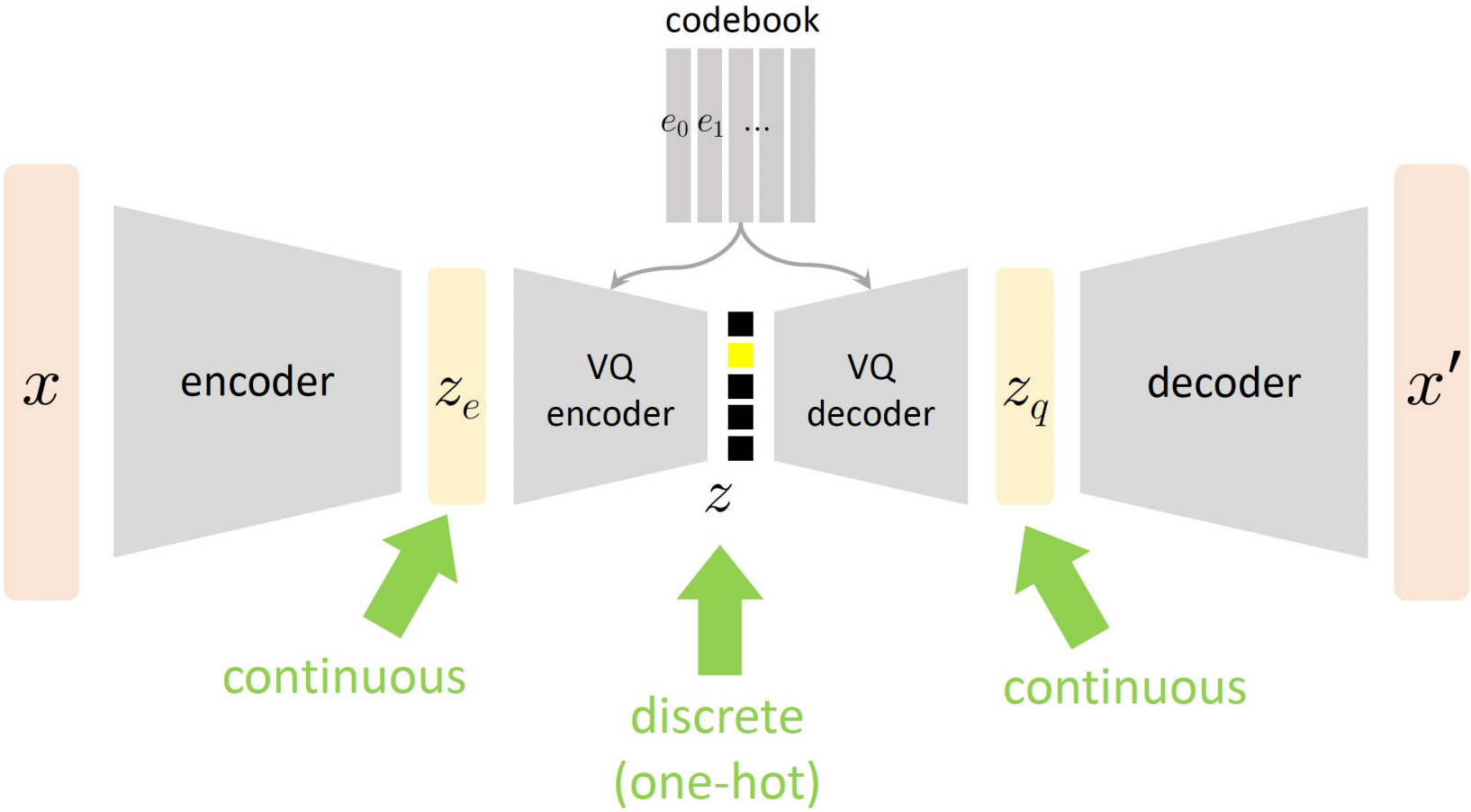
Advanced Variants of VAE

- **Discrete Latent Variable Models**
  - **Motivation:** Train via Maximum Likelihood ➔ Maximizing ELBO
    - ✓ Reconstruction Loss + Regularization Loss



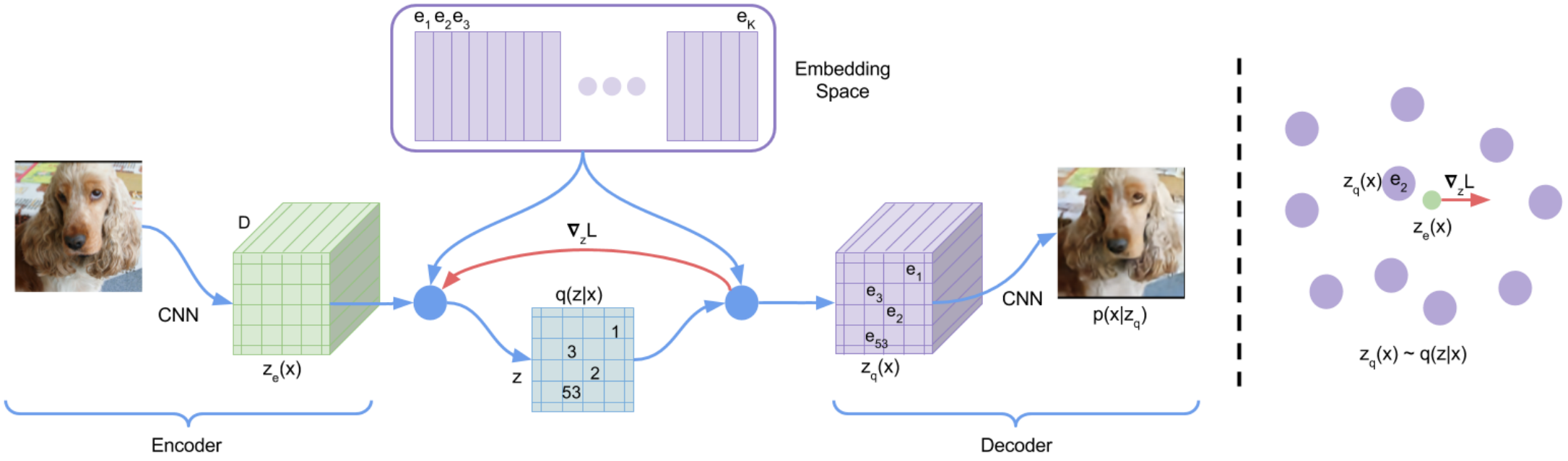
Advanced Variants of VAE

- **Vector-Quantized VAE (VQ-VAE)**
  - **Motivation: Indexing** *learned* continuous feature with *learned* codebook



Advanced Variants of VAE

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  - ➢ **Motivation: Indexing** *learned* continuous feature with *learned* codebook
    - ➔ Similar to the unsupervised clustering method: **K-Means**

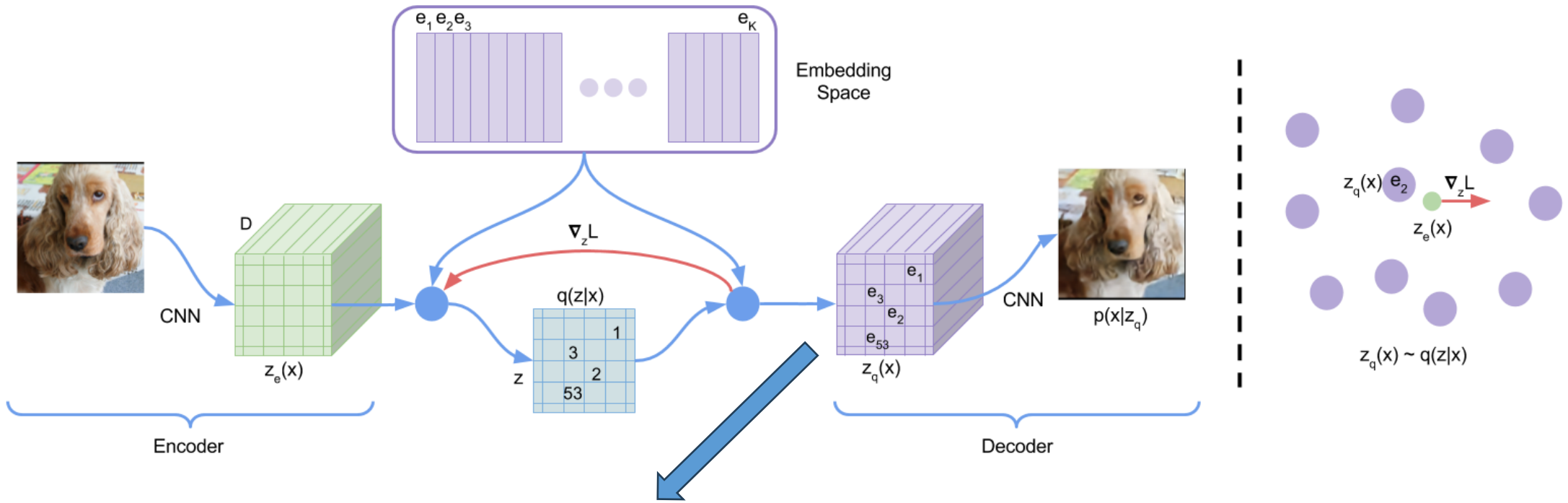


Discrete latent code (index of embedding)

Oord et al., Neural Discrete Representation Learning, NeurIPS, 2017.

Advanced Variants of VAE

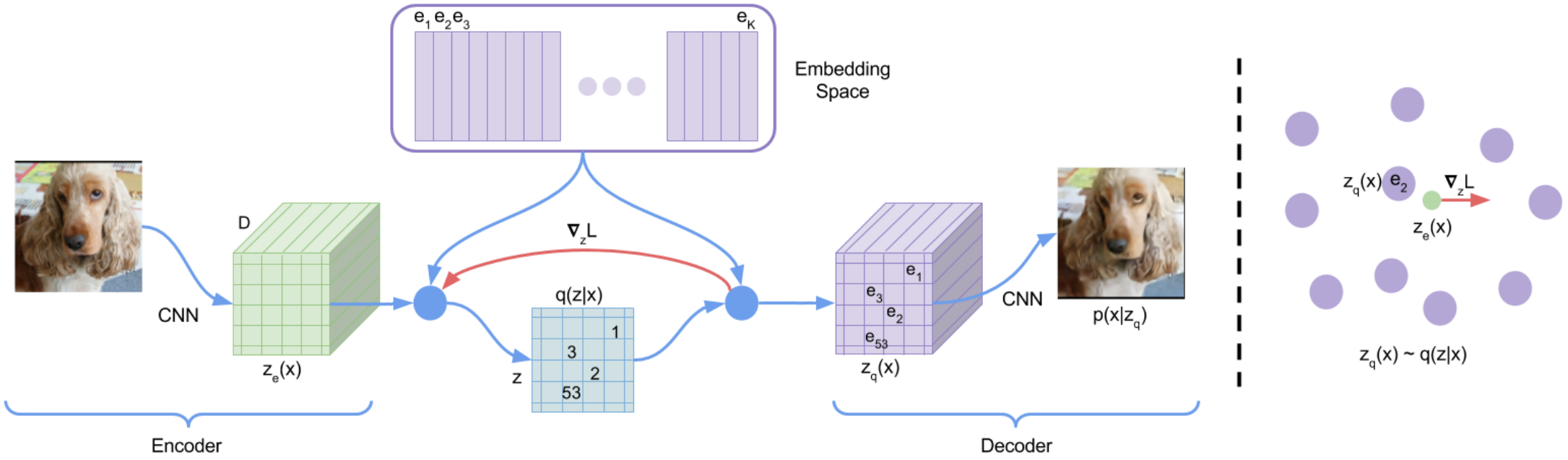
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$$\mathbf{z}_q(\mathbf{x}) = \text{Quantize}(E(\mathbf{x})) = \mathbf{e}_k \text{ where } k = \arg\min_i \|E(\mathbf{x}) - \mathbf{e}_i\|_2$$

Advanced Variants of VAE

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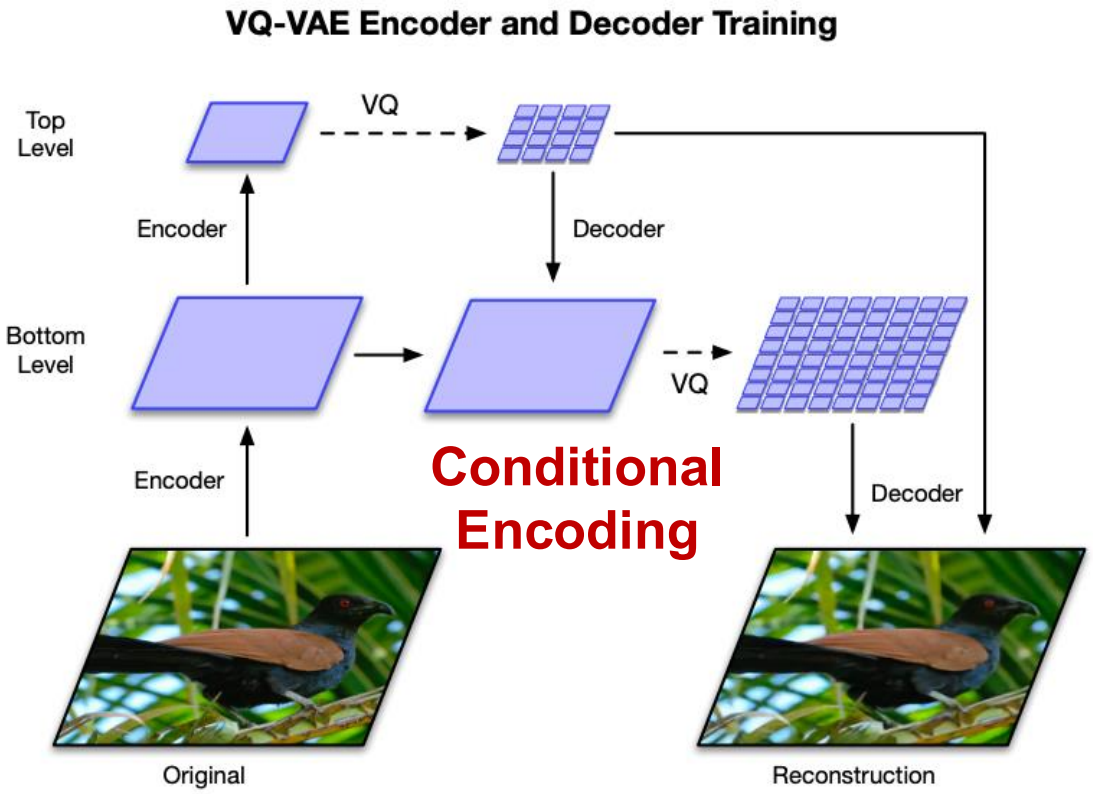


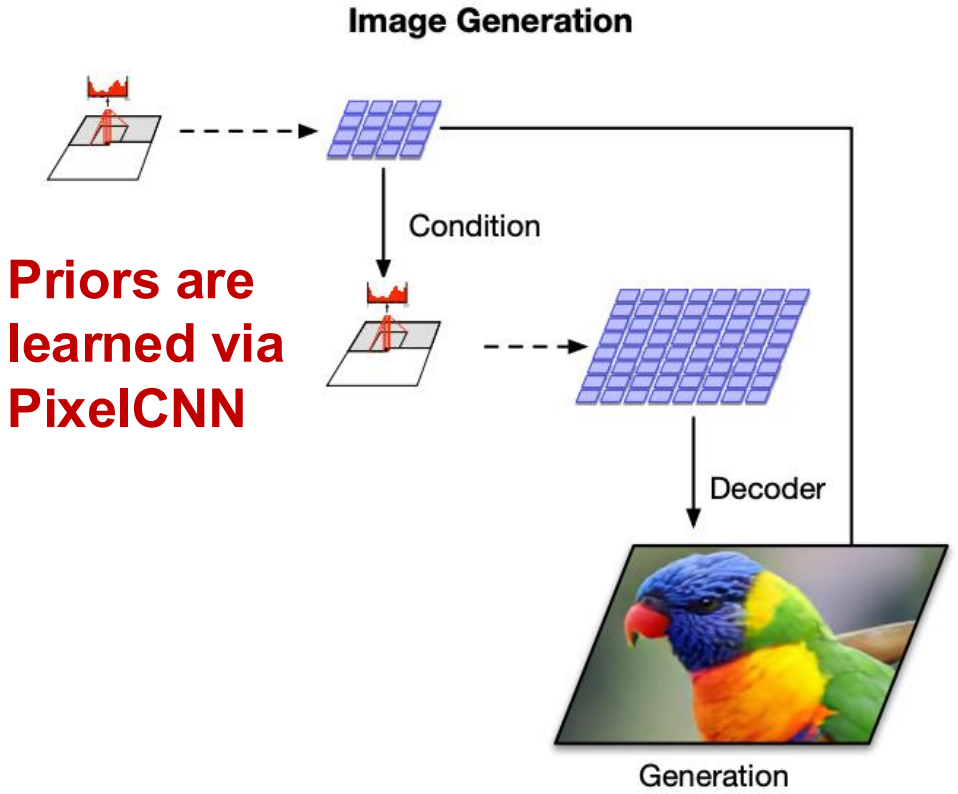
$$L = \underbrace{\|\mathbf{x} - D(\mathbf{e}_k)\|_2^2}_{\text{reconstruction loss}} + \underbrace{\|\text{sg}[E(\mathbf{x})] - \mathbf{e}_k\|_2^2}_{\text{VQ loss}} + \underbrace{\beta\|E(\mathbf{x}) - \text{sg}[\mathbf{e}_k]\|_2^2}_{\text{commitment loss}}$$

sg[] is the stop-gradient operator

Advanced Variants of VAE

- **Learning Discrete Latent Code: VQ-VAE2**
  - **Motivation:** Learn a multi-layer hierarchical VQ-VAE to separately capture local patterns and global information





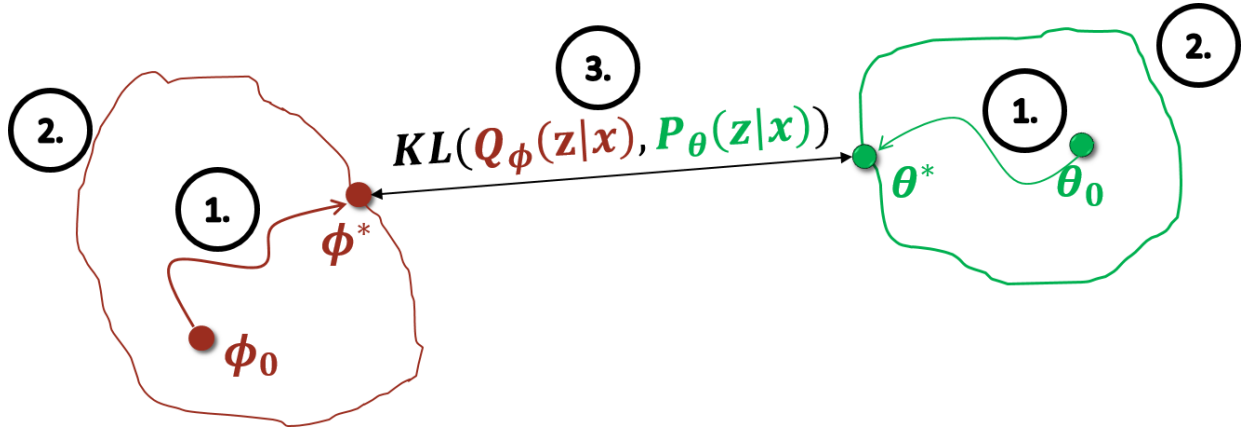
A. Razavi et al., Generating Diverse High-Fidelity Images with VQ-VAE-2, NeurIPS, 2019.

Advanced Variants of VAE

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Advanced Variants of VAE

- **Research Directions in VAE**



① **Better Optimization Techniques**

✓ Improved Amortization

② **More expressive approximating families**

✓ More complex variational posterior $q_\phi(z \mid x)$, learned priors $p(z; \psi)$

③ **Alternate loss functions**

✓ Maximum Mean Discrepancy (MMD), Wasserstein distance

Summary

- **Latent Variable Models**
  - **Pros**
    - ✓ Naturally discover the latent structure of data (unsupervised)
    - ✓ Latent representations for any x can be inferred via $q_\phi(z \mid x)$
    - ✓ Combine multiple simple models $p_\theta(x \mid z)$ to get a more flexible one
    - ✓ Allow efficient generation $z \sim p(z)$, $x \sim p_\theta(x \mid z)$
  - **Cons**
    - ✓ Log-likelihood $\log p_\theta(x)$ is intractable, thus learning is difficult
    - ✓ Optimize the tractable ELBO instead, rely on joint learning of a model (θ) and an amortized inference component (ϕ)

Supplementary Material

- **Stanford CS 236 Lecture 6**
  - https://www.youtube.com/watch?v=8cO61e_8oPY&list=PLoROMvodv4rPOWA-omMM6STXaWW4FvJT8&index=6
- **Cornell CS 6785 Lecture 6**
  - https://www.youtube.com/watch?v=v86nlzoYrYI&list=PL2UML_KCiC0UPzjW9BjO-lW6dqliu9O4B&index=6
- **Deep Generative Models**
  - Chapter 5 Latent Variable Models
- **The Principle of Diffusion Models**
  - Chapter B.2 Variational Perspective: From VAEs to DDPMs

Supplementary Material

- **Reading List**
  - Kingma et al., Auto-Encoding Variational Bayes, ICLR, 2014
  - Rezende et al., Stochastic Backpropagation and Approximate Inference in Deep Generative Models, ICML, 2014
  - Burda et al., Importance Weighted Autoencoders, ICLR, 2016
  - Kingma et al., Improving Variational Inference with Inverse Autoregressive Flow, NeurIPS, 2016
  - Higgins et al., beta-VAE: Learning Basic Visual Concepts with a Constrained Variational Framework, ICLR, 2017
  - Bowman et al., Generating Sentences from a Continuous Space, CoNLL, 2016
  - van den Oord et al., Neural Discrete Representation Learning, NeurIPS, 2017
  - He et al., Lagging Inference Networks and Posterior Collapse in Variational Autoencoders, ICLR, 2019